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Kim et al.

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(54) **DISPLAY APPARATUS**

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(30) **Foreign Application Priority Data**

Feb. 16, 2023 (KR) 10-2023-0020904

(51) **Int. Cl.**
G09G 3/20 (2006.01)
G09G 3/32 (2016.01)

(52) **U.S. Cl.**
CPC ***G09G 3/32*** (2013.01); *G09G 2310/027* (2013.01); *G09G 2320/0276* (2013.01); *G09G 2330/021* (2013.01)

(58) **Field of Classification Search**
CPC G09G 3/32; G09G 2310/027; G09G 2320/0276; G09G 2330/021
USPC 345/55
See application file for complete search history.

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(57) **ABSTRACT**

A display apparatus includes: a gamma reference voltage generator, a digital-to-analog converter, an amplifier and a bias voltage applier. The gamma reference voltage generator is configured to generate a gamma reference voltage. The digital-to-analog converter is configured to convert a data signal to a data voltage based on the gamma reference voltage. The amplifier is configured to receive the data voltage from the digital-to-analog converter and to output the data voltage to a data line. The bias voltage applier is configured to output a first bias voltage to the gamma reference voltage generator and to output a second bias voltage to the amplifier. The bias voltage applier is turned off while input image data have a black image.

20 Claims, 17 Drawing Sheets

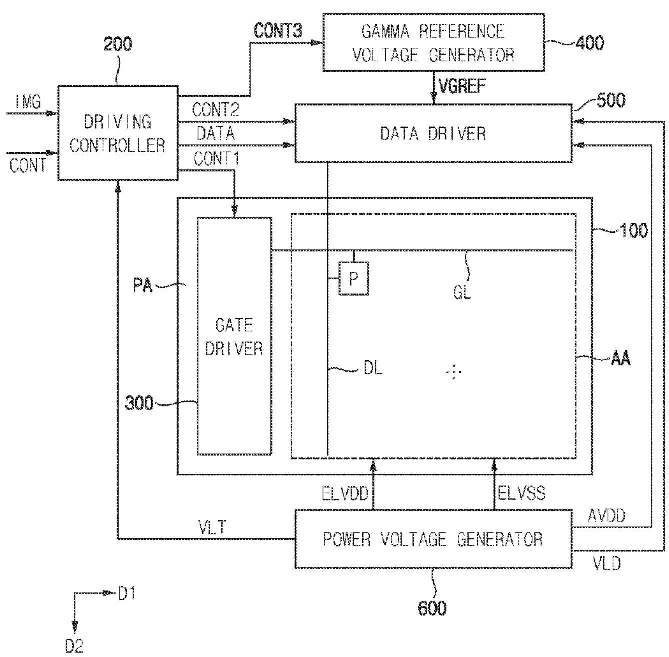
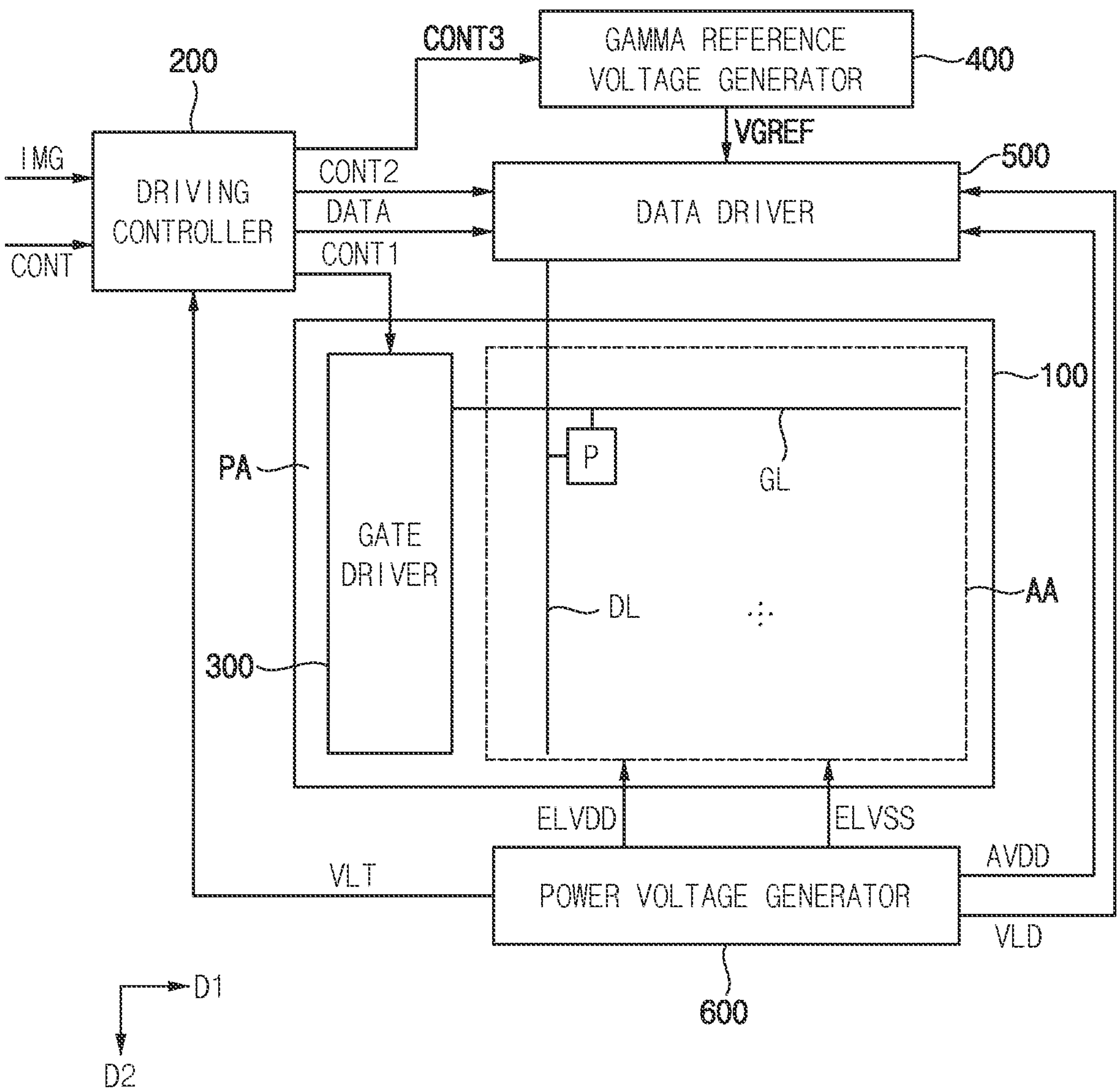
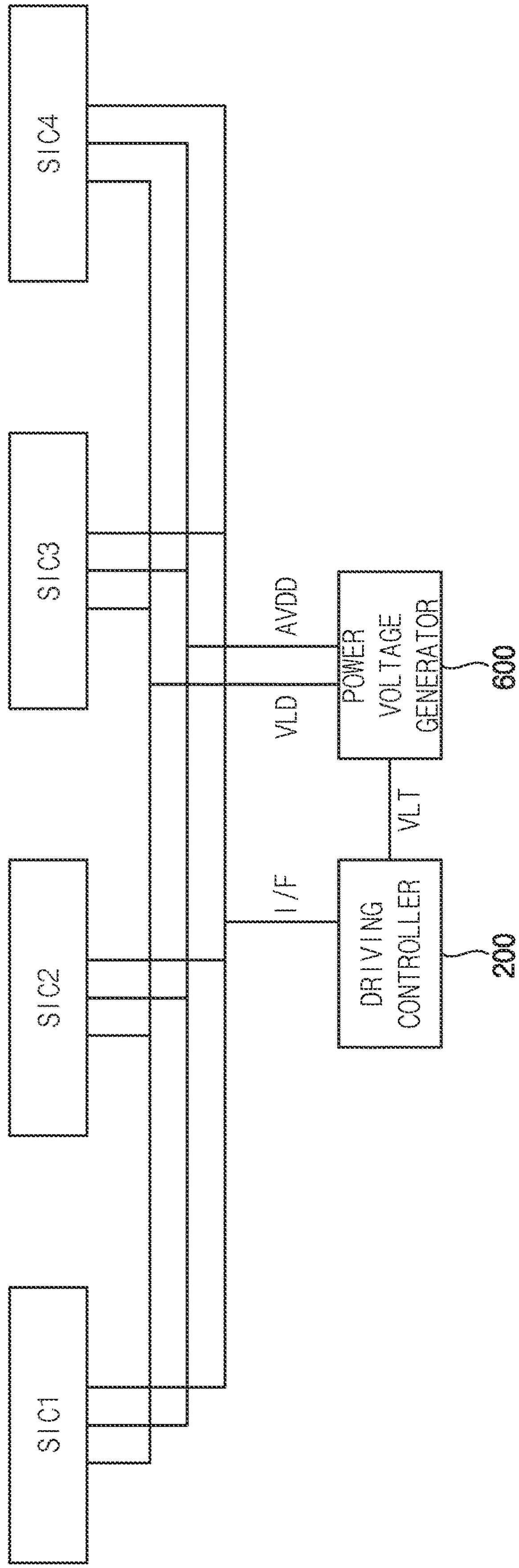


FIG. 1



200
IMG
CONT
DRIVING CONTROLLER
CONT3
GAMMA REFERENCE VOLTAGE GENERATOR
400
VGREF
500
CONT2
DATA
CONT1
DATA DRIVER
100
PA
P
GL
GATE DRIVER
DL
AA
300
ELVDD
ELVSS
AVDD
VLT
POWER VOLTAGE GENERATOR
VLD
600
D1
D2

FIG. 2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SIC2
SIC1
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POWER VOLTAGE GENERATOR
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VLT
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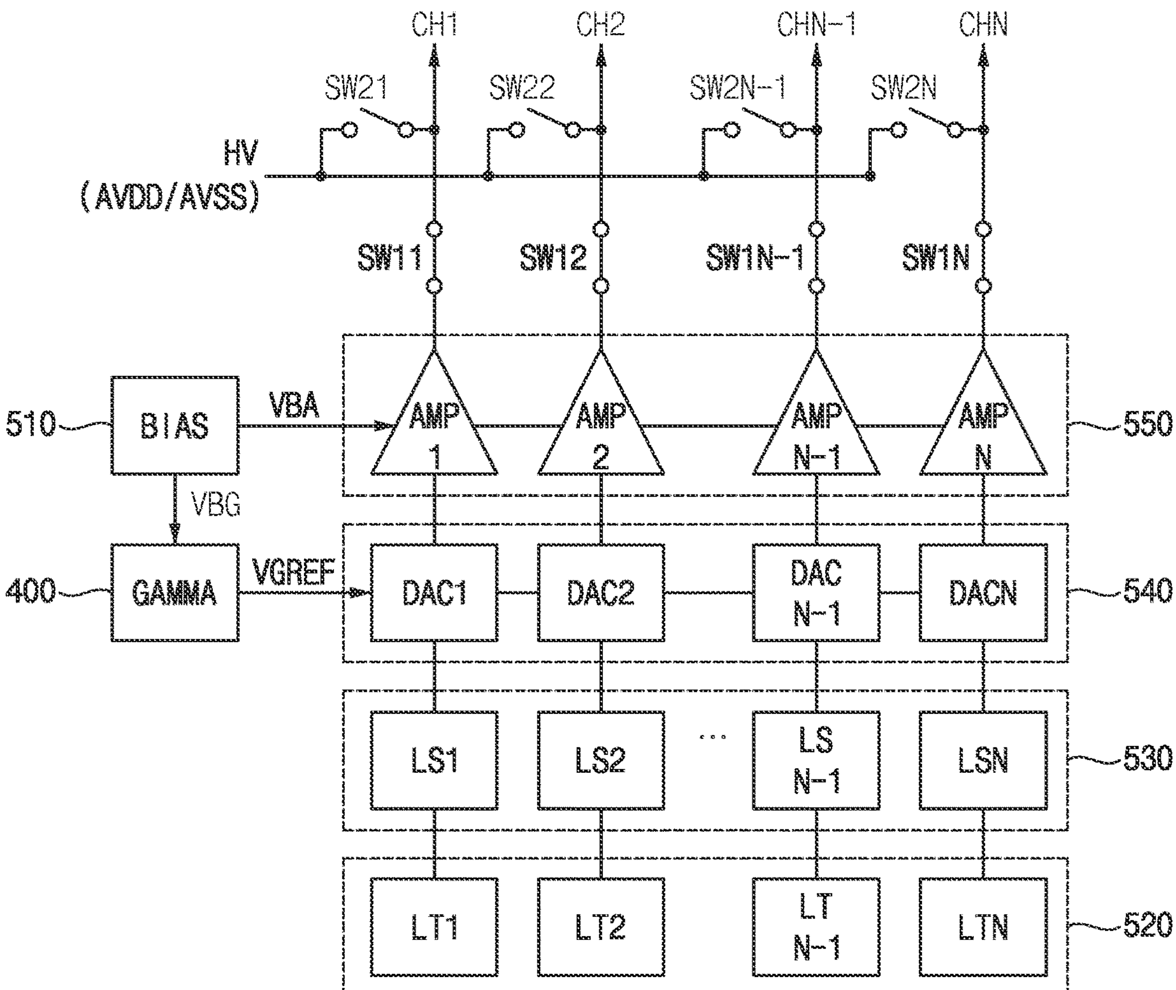
DRIVING CONTROLLER
200

FIG. 3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AMP 2
AMP N-1
AMP N
550
VBG
400
GAMMA
VGREF
DAC1
DAC2
DAC N-1
DACN
540
LS1
LS2
...
LS N-1
LSN
530
LT1
LT2
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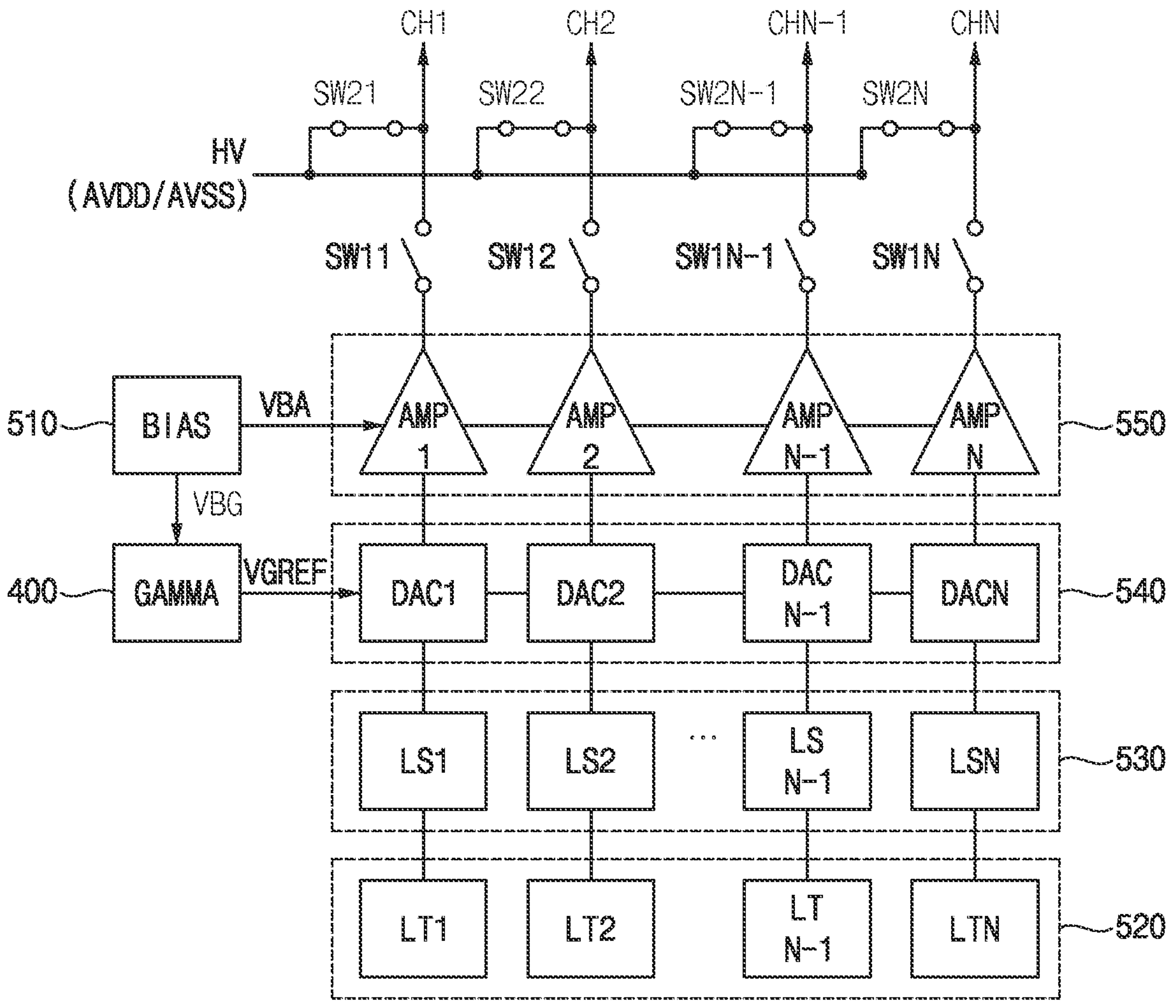
LTN
520

FIG. 4



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AMP 2
AMP N-1
AMP N
550
VBG
400
GAMMA
VGREF
DAC1
DAC2
DAC N-1
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LSN
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LT1
LT2
LT N-1
LTN
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FIG. 5

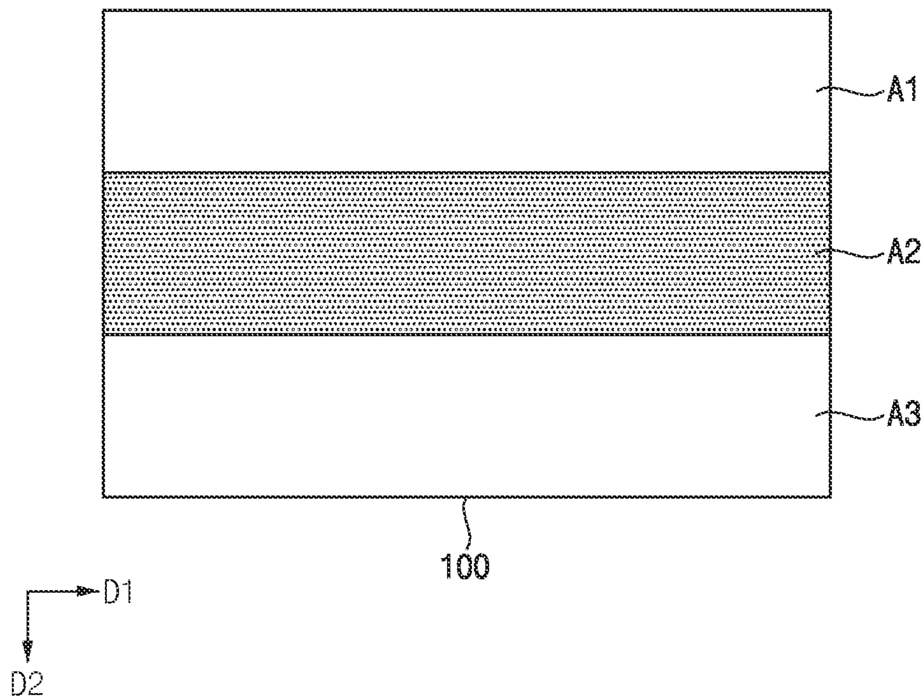
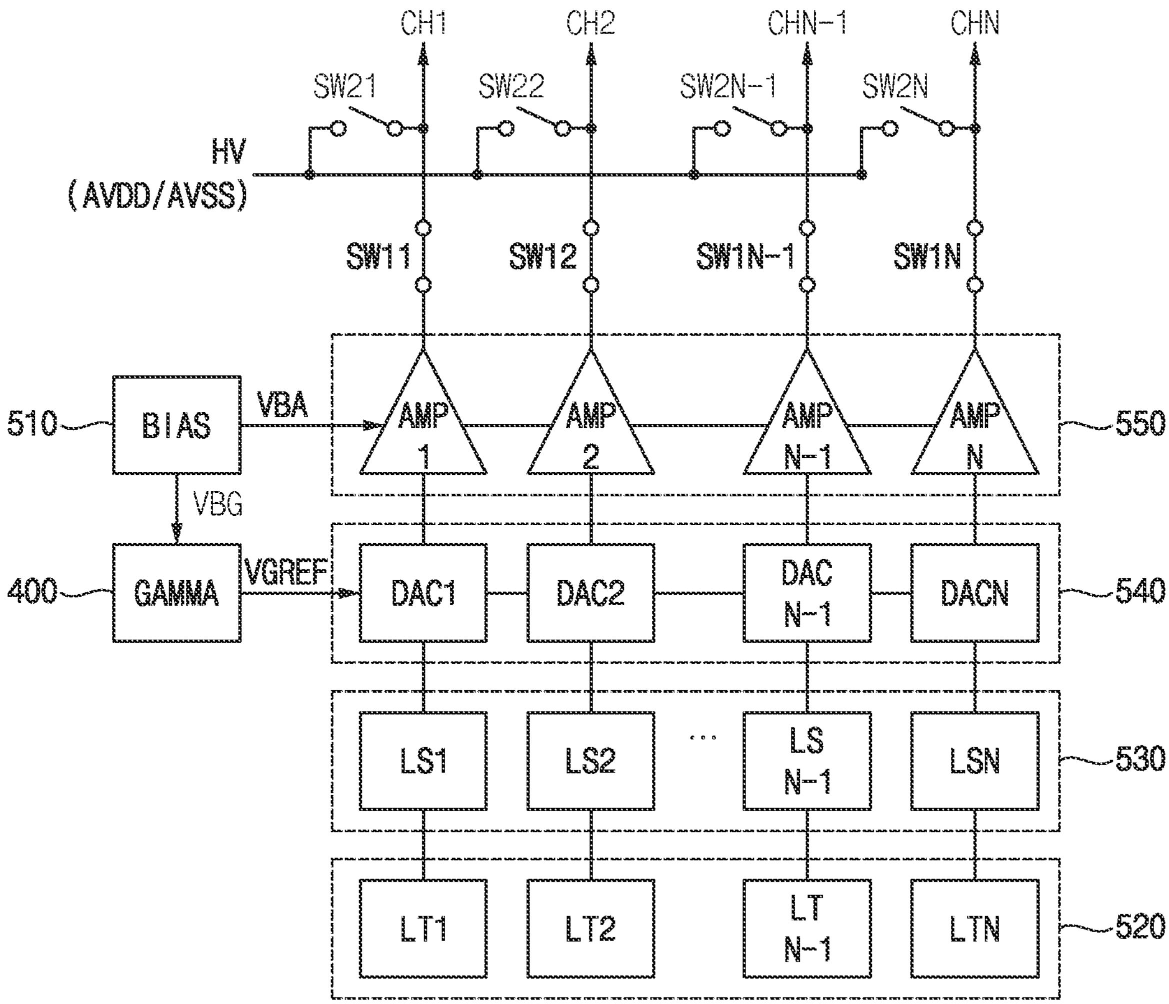


FIG. 6

FIG. 7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AMP N
550
VBG
400
GAMMA
VGREF
DAC1
DAC2
DAC N-1
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LS2
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LSN
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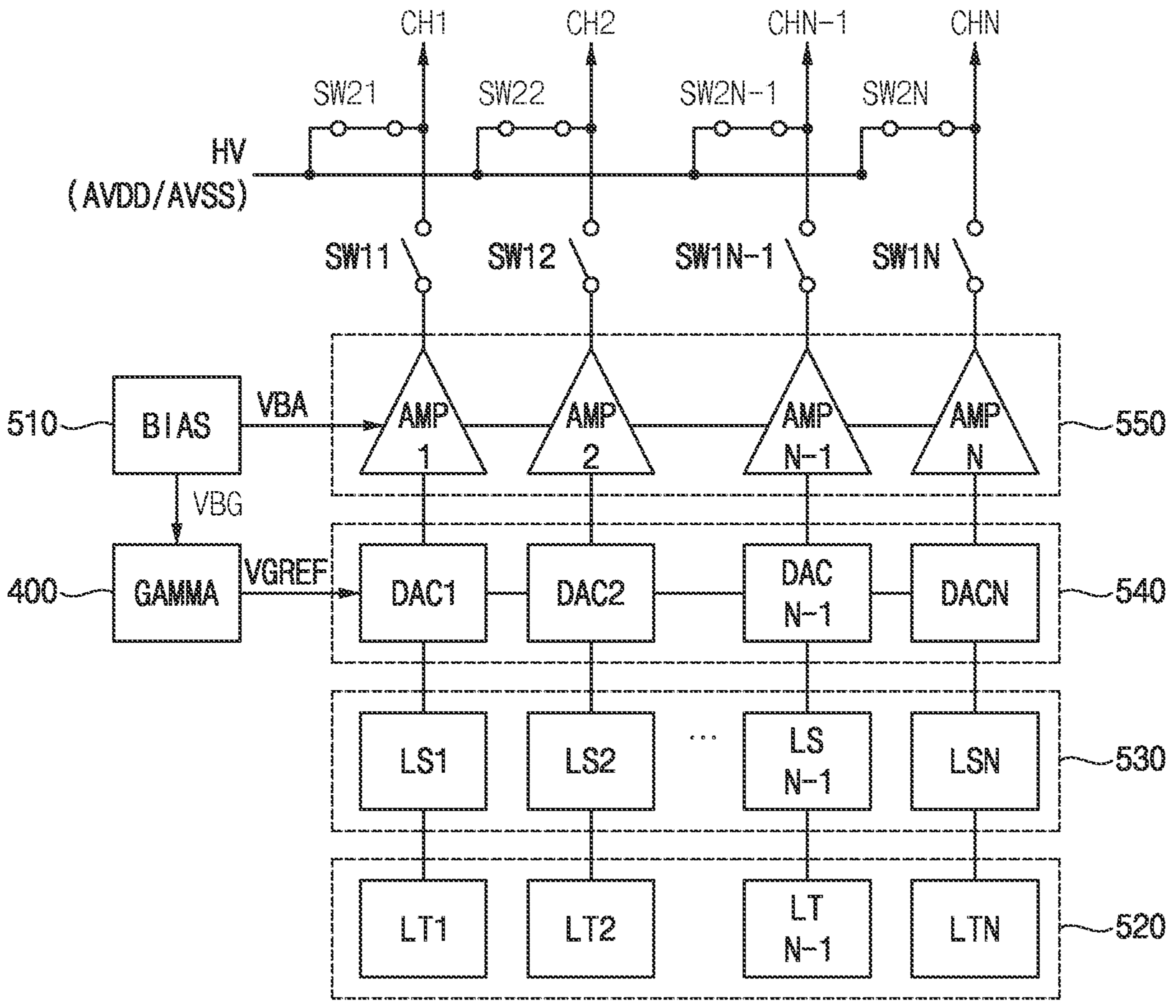
LT1
LT2
LT N-1
LTN
520

FIG. 8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VBA
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AMP 2
AMP N-1
AMP N
550
VBG
400
GAMMA
VGREF
DAC1
DAC2
DAC N-1
DACN
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LS1
LS2
...
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LT1
LT2
LT N-1
LTN
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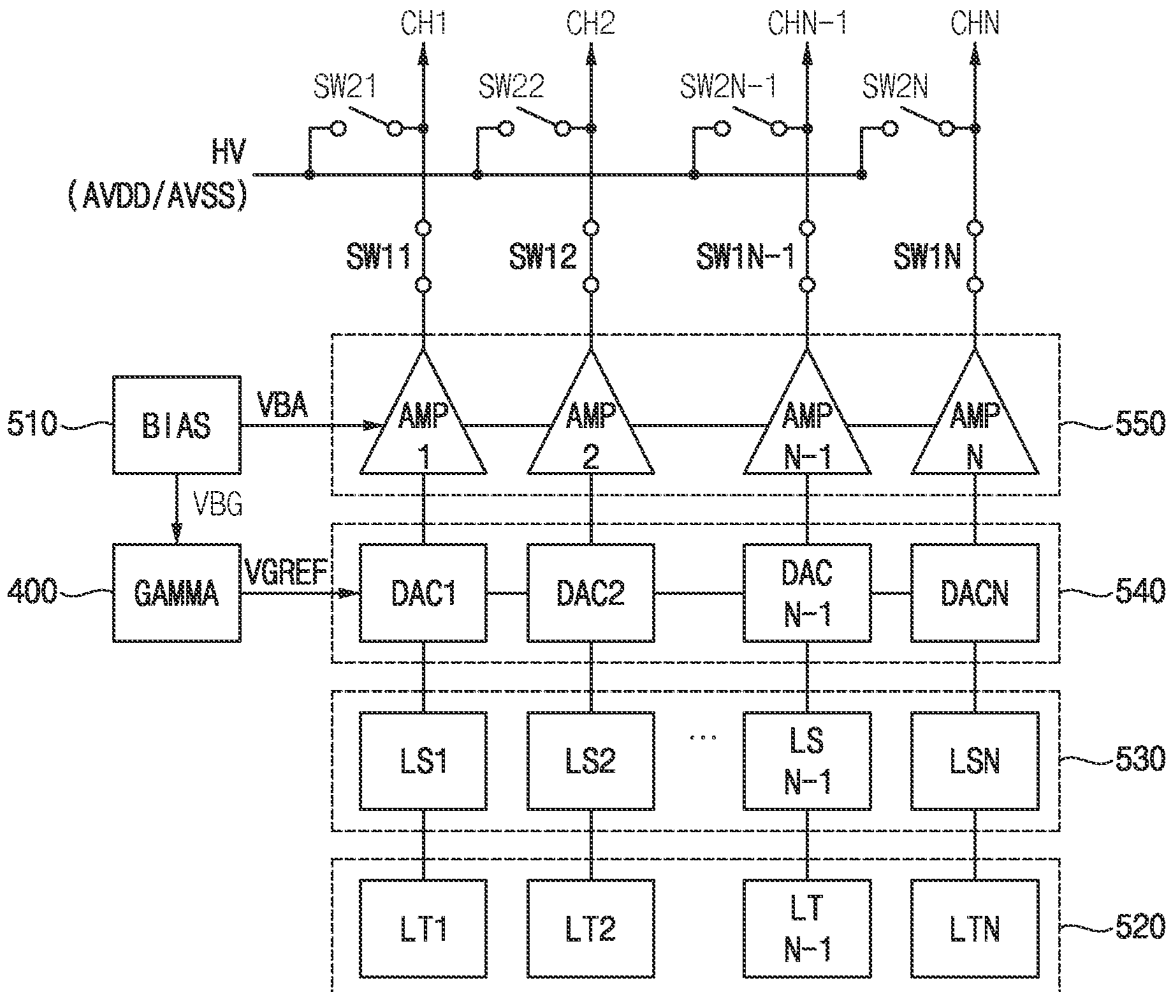
FIG. 9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AMP N
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VGREF
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LTN
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FIG. 10

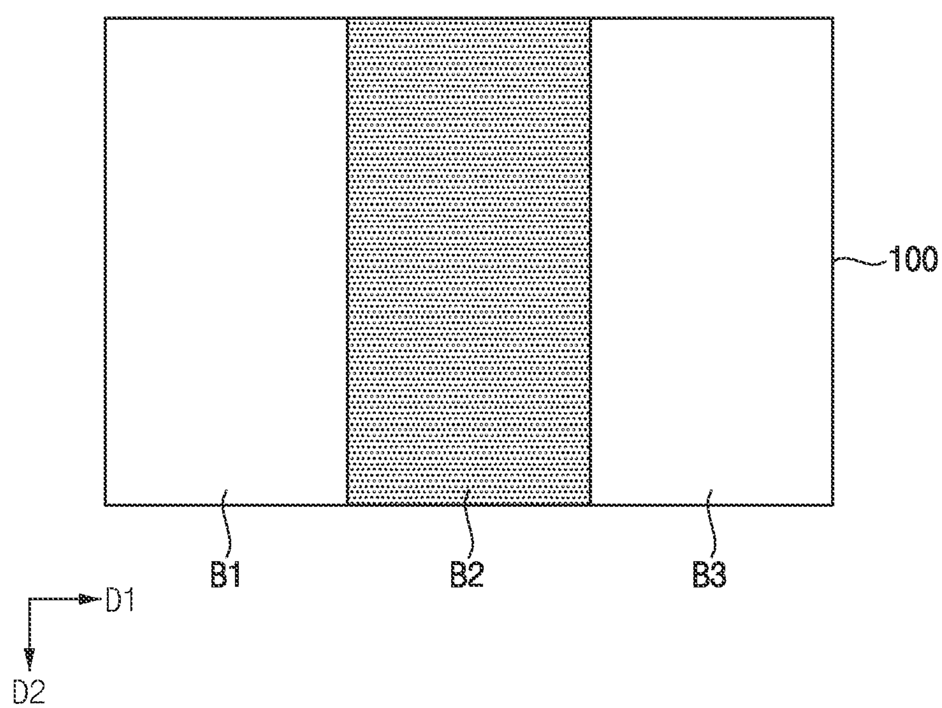
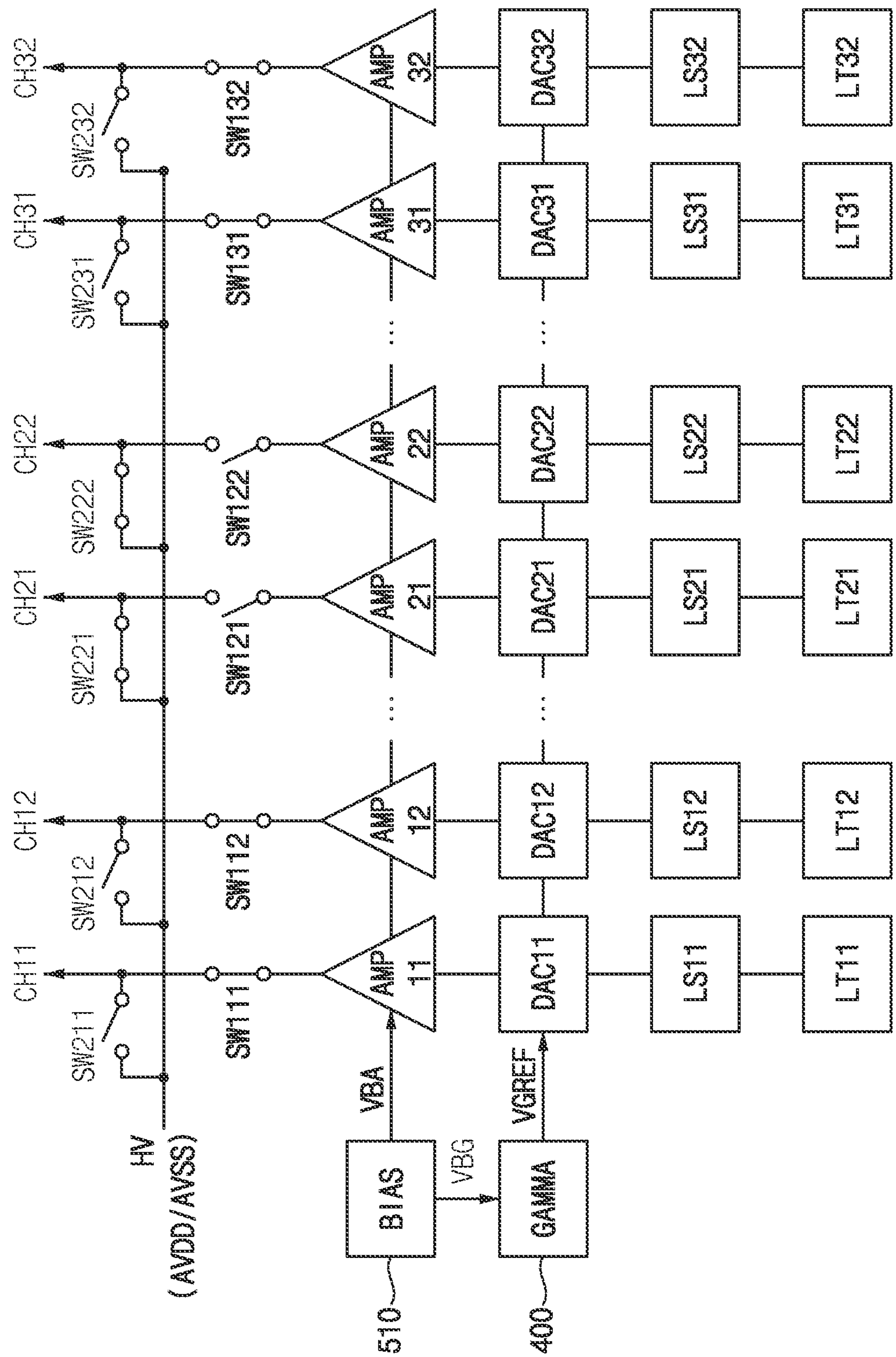


FIG. 11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SW131
SW132
AMP 11
AMP 12
AMP 21
AMP 22
AMP 31
AMP 32
DAC11
DAC12
DAC21
DAC22
DAC31
DAC32
LS11
LS12
LS21
LS22
LS31
LS32
LT11
LT12
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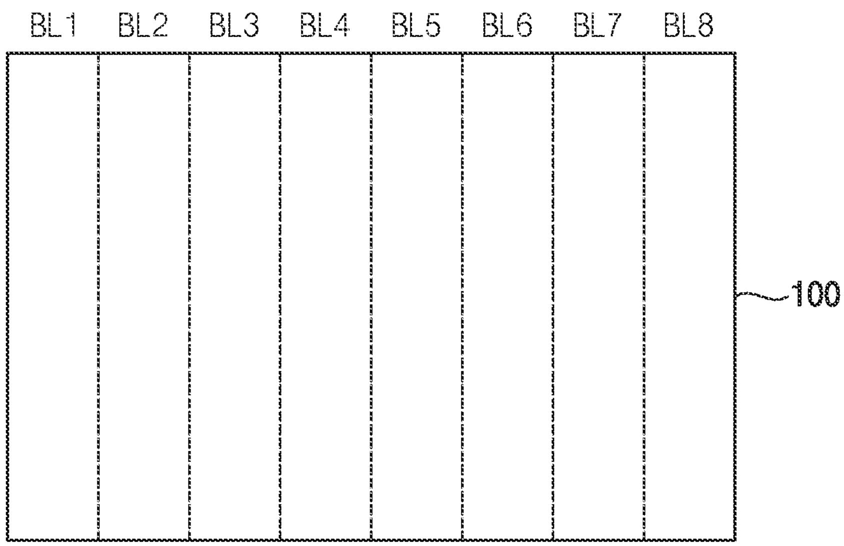
510
400

FIG. 12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BL6
BL7
BL8
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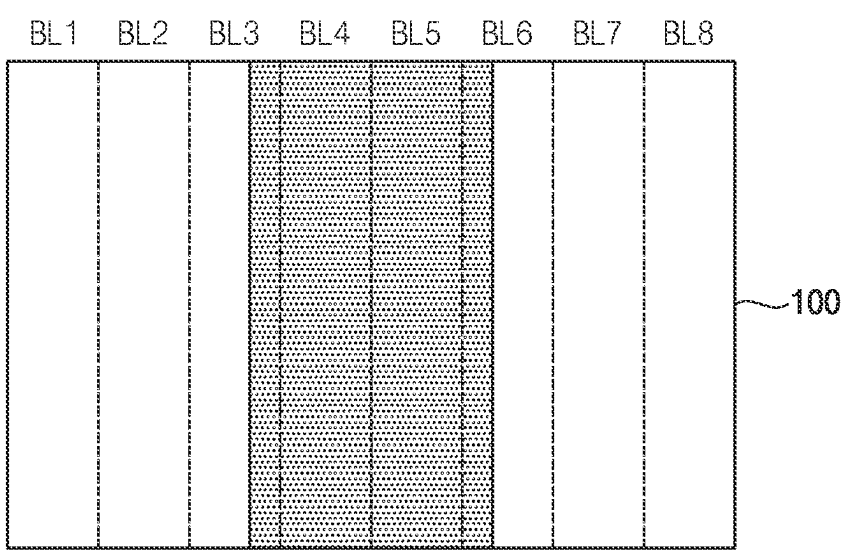
FIG. 13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BL8
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FIG. 14

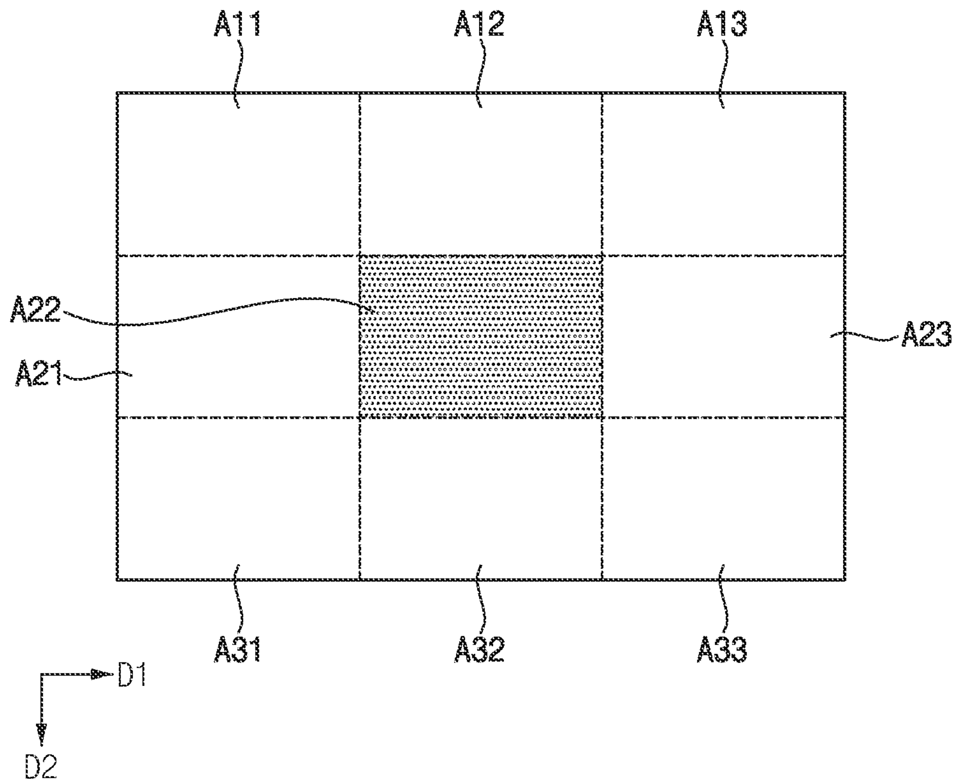
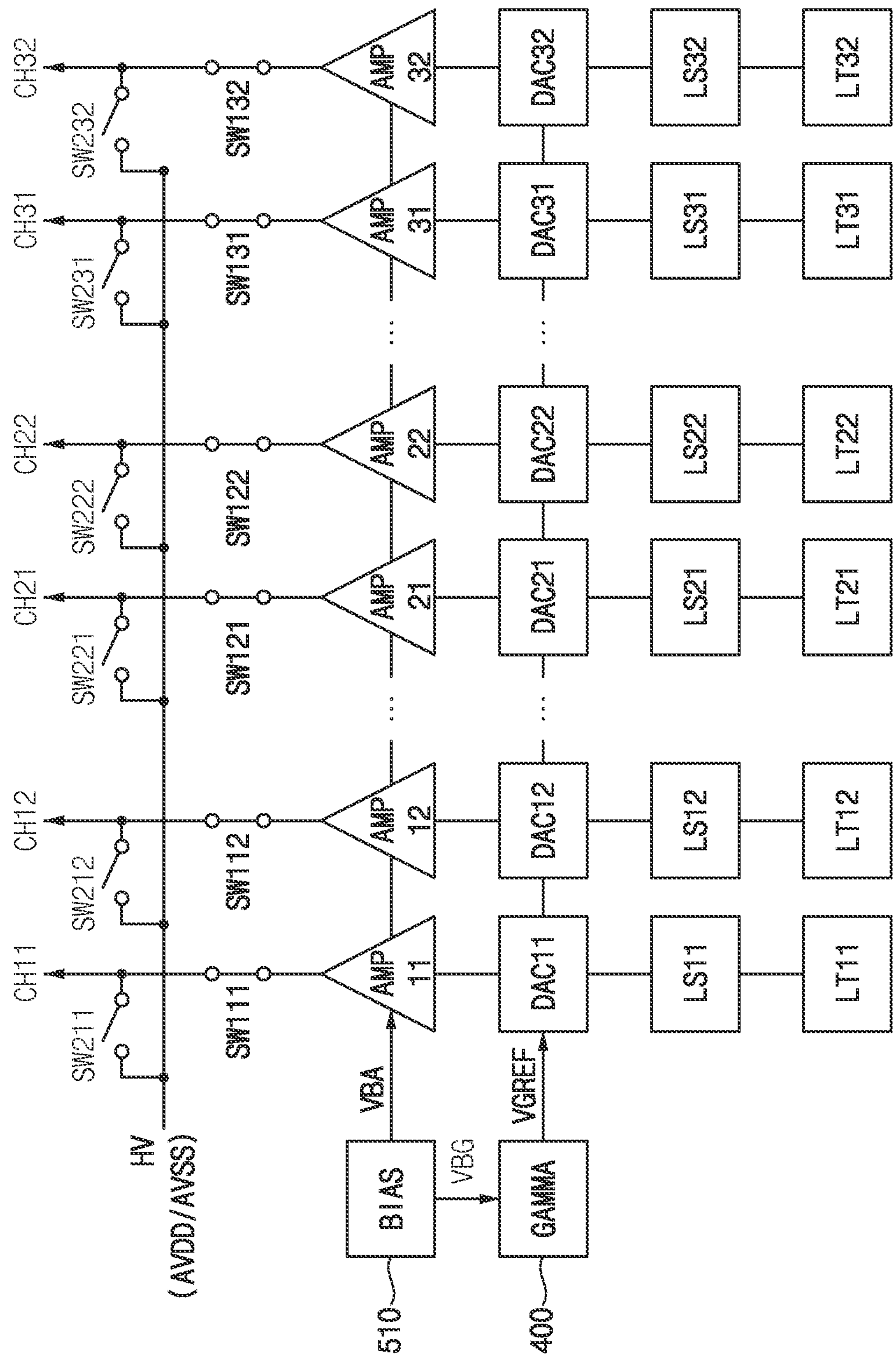


FIG. 15



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DAC21
DAC22
DAC31
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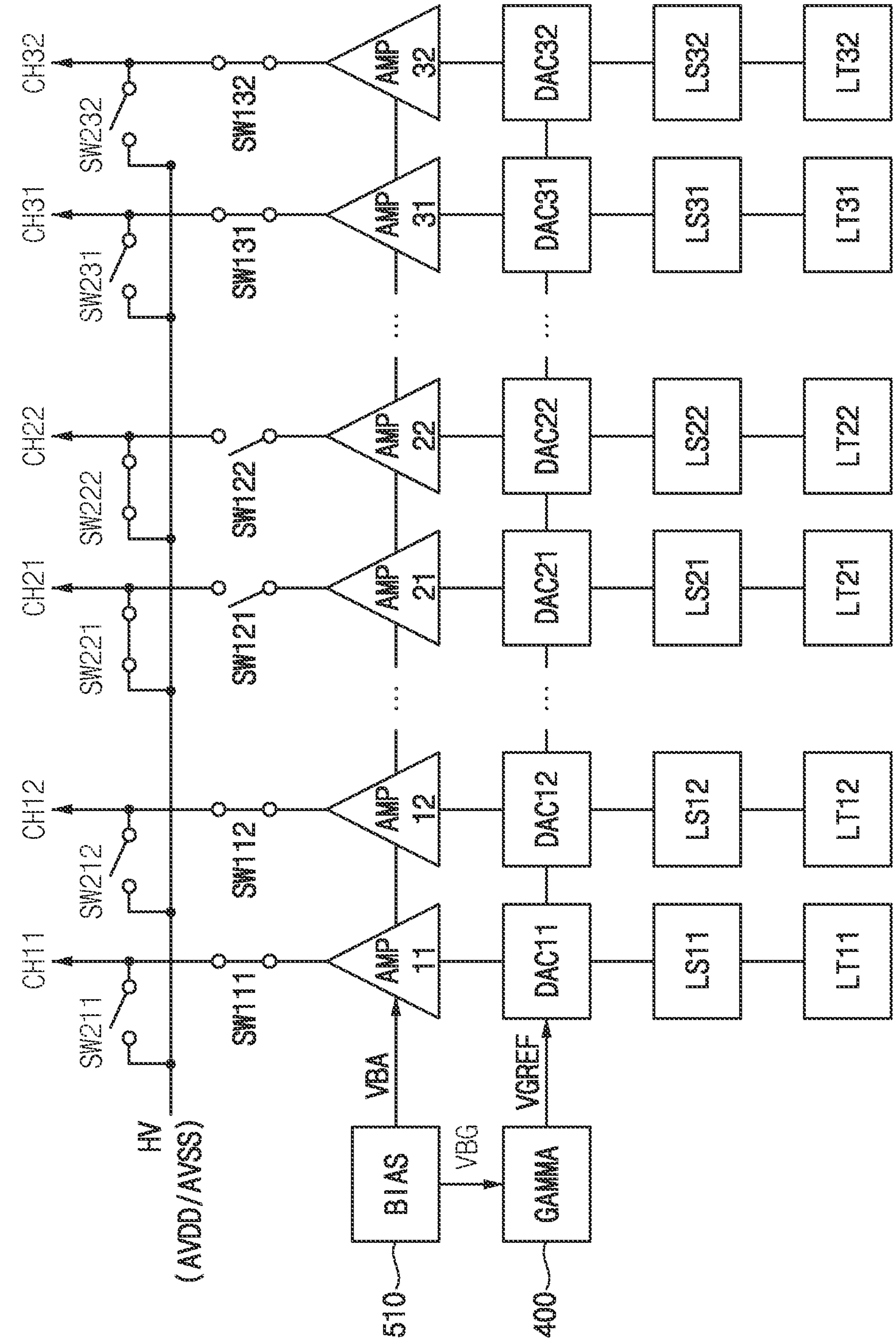
LT32
510
400

FIG. 16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DAC12
DAC21
DAC22
DAC31
DAC32
LS11
LS12
LS21
LS22
LS31
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LT32
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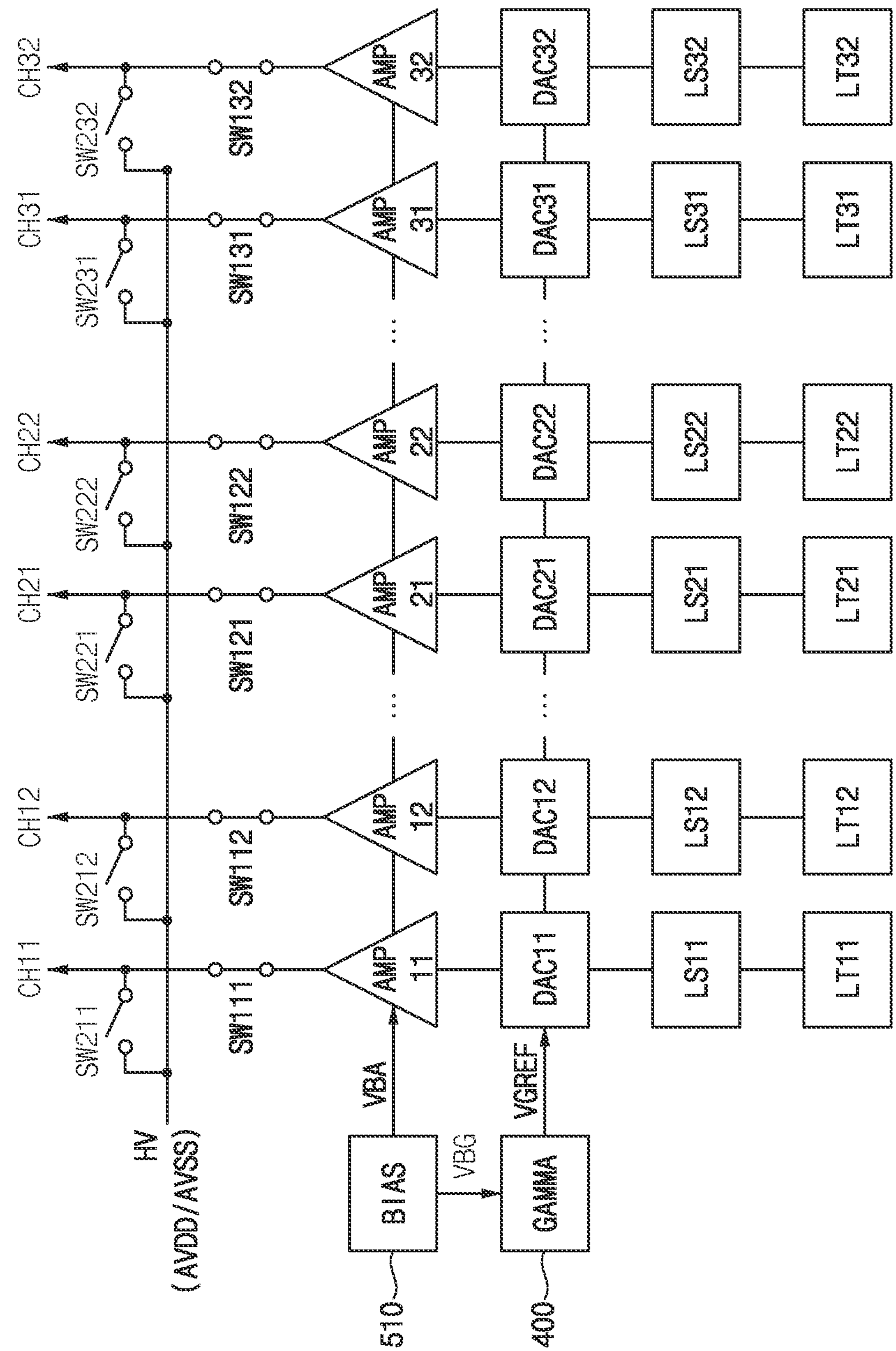
VBG
VGREF
BIAS
GAMMA
510
400

FIG. 17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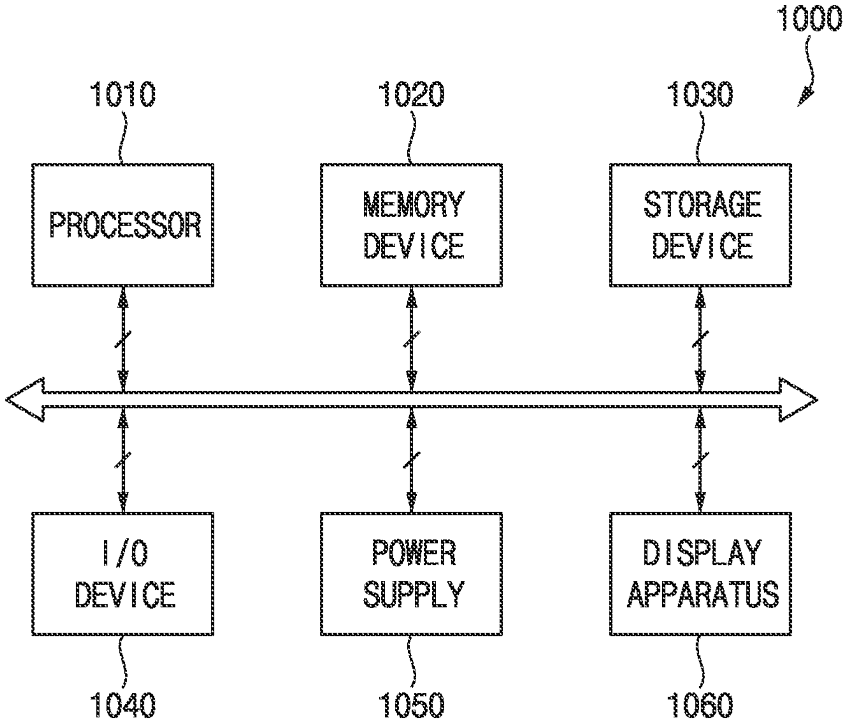
LT32
510
400

FIG. 18


1000
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1020
1030
PROCESSOR
MEMORY DEVICE
STORAGE DEVICE
I/O DEVICE
POWER SUPPLY
DISPLAY APPARATUS
1040
1050
1060

FIG. 19

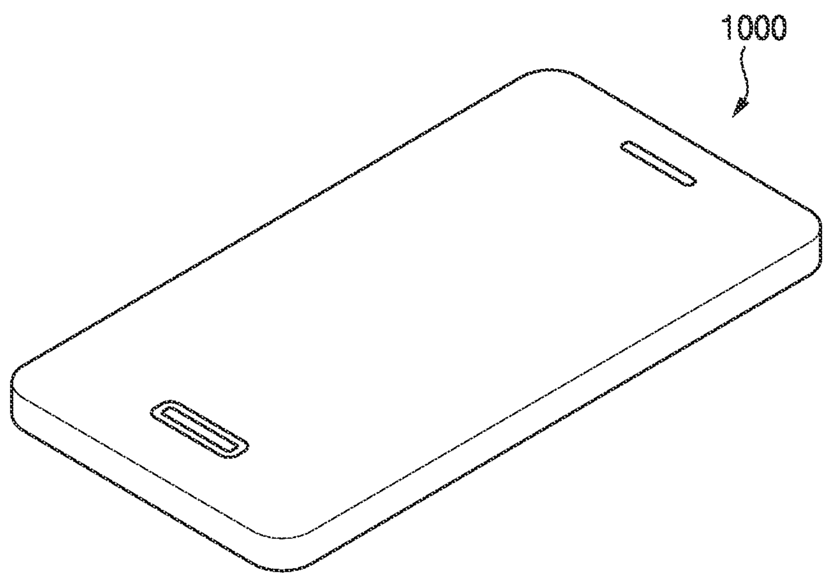
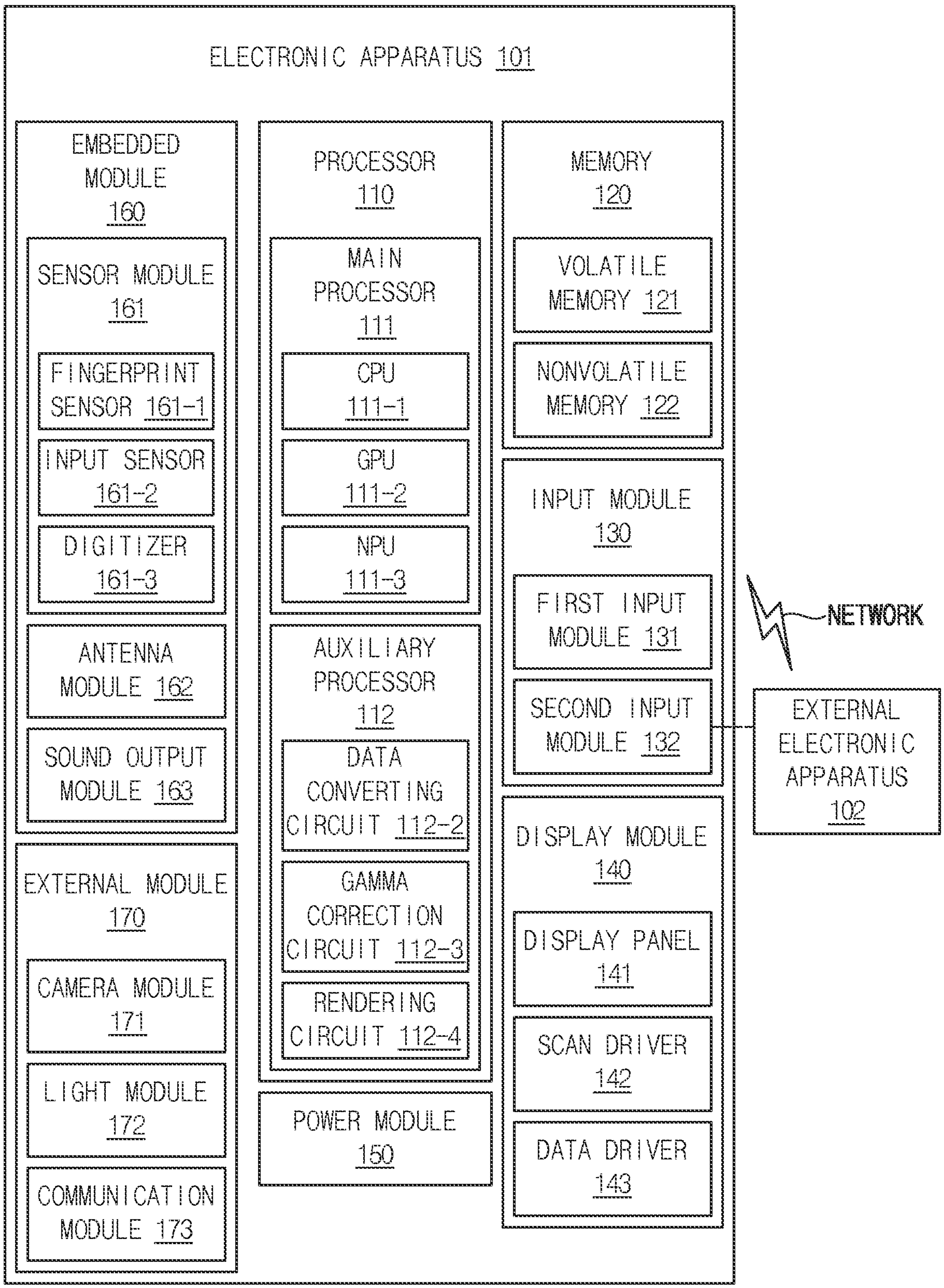

1000

FIG. 20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EMBEDDED MODULE 160
SENSOR MODULE 161
FINGERPRINT SENSOR 161-1
INPUT SENSOR 161-2
DIGITIZER 161-3
ANTENNA MODULE 162
SOUND OUTPUT MODULE 163
EXTERNAL MODULE 170
CAMERA MODULE 171
LIGHT MODULE 172
COMMUNICATION MODULE 173
PROCESSOR 110
MAIN PROCESSOR 111
CPU 111-1
GPU 111-2
NPU 111-3
AUXILIARY PROCESSOR 112
DATA CONVERTING CIRCUIT 112-2
GAMMA CORRECTION CIRCUIT 112-3
RENDERING CIRCUIT 112-4
POWER MODULE 150
MEMORY 120
VOLATILE MEMORY 121
NONVOLATILE MEMORY 122
INPUT MODULE 130
FIRST INPUT MODULE 131
SECOND INPUT MODULE 132
DISPLAY MODULE 140
DISPLAY PANEL 141
SCAN DRIVER 142
DATA DRIVER 143
NETWORK
EXTERNAL ELECTRONIC APPARATUS 102

DISPLAY APPARATUS

This application claims priority to Korean Patent Application No. 10-2023-0020904, filed on Feb. 16, 2023, and all the benefits accruing therefrom under 35 U.S.C. § 119, the content of which in its entirety is herein incorporated by reference.

BACKGROUND

1. Field

Embodiments of the present invention relate to a display apparatus. More particularly, embodiments of the present invention relate to a display apparatus reducing a power consumption of a data driver.

2. Description of the Related Art

Generally, a display apparatus includes a display panel and a display panel driver. The display panel displays an image based on input image data. The display panel includes a plurality of gate lines, a plurality of data lines and a plurality of pixels. The display panel driver includes a gate driver and a data driver. The gate driver outputs gate signals to the gate lines. The data driver outputs data voltages to the data lines. The display panel driver may further include a gamma reference voltage generator providing a gamma reference voltage to the data driver.

High constant current may flow through amplifiers of the gamma reference voltage generator and the data driver so that a power consumption of the display apparatus may be high.

SUMMARY

Embodiments of the present invention provide a display apparatus capable of reducing a power consumption of the display apparatus by operating a gamma reference voltage generator and a data driver in a black mode when a display panel displays a black image.

In an embodiment of a display apparatus according to the present invention, the display apparatus includes: a gamma reference voltage generator, a digital-to-analog converter, an amplifier and a bias voltage applier. The gamma reference voltage generator is configured to generate a gamma reference voltage. The digital-to-analog converter is configured to convert a data signal to a data voltage based on the gamma reference voltage. The amplifier is configured to receive the data voltage from the digital-to-analog converter and to output the data voltage to a data line. The bias voltage applier is configured to output a first bias voltage to the gamma reference voltage generator and to output a second bias voltage to the amplifier. The bias voltage applier is turned off while input image data have a black image.

In an embodiment, the gamma reference voltage generator and the amplifier may be turned off while the input image data have the black image.

In an embodiment, the display apparatus may further include: a channel connected to the data line, a first switch connected between an output terminal of the amplifier and the channel and a second switch connected between an input terminal of a hold voltage and the channel.

In an embodiment, in a normal mode in which the input image data include a grayscale value which is not the black image, the first switch may be turned on and the second switch may be turned off.

In an embodiment, in a black mode in which the input image data include only the black image, the first switch may be turned off and the second switch may be turned on.

In an embodiment, the hold voltage may be a high data power voltage generated by a power voltage generator.

In an embodiment of a display apparatus according to the present invention, the display apparatus includes: a display panel, a channel, an amplifier, a first switch and a second switch. The display panel includes a data line extending in a second direction. The channel is connected to the data line. The amplifier is configured to output a data voltage. The first switch is connected between an output terminal of the amplifier and the channel. The second switch is connected between an input terminal of a hold voltage and the channel. When input image data have a first black pattern extending in a first direction perpendicular to the second direction, the first switch is turned off and the second switch is turned on during a black period corresponding to the first black pattern, and the first switch is turned on and the second switch is turned off during a normal period not corresponding to the first black pattern.

In an embodiment, a data signal may include a line configuration and line data. The line configuration may include a setting signal representing the black period and the normal period.

In an embodiment, when proceeding from the black period to the normal period, the setting signal may be determined based on a wakeup time.

In an embodiment, the display apparatus may further include a gamma reference voltage generator configured to generate a gamma reference voltage, a digital-to-analog converter configured to convert a data signal to the data voltage based on the gamma reference voltage and a bias voltage applier configured to output a first bias voltage to the gamma reference voltage generator and to output a second bias voltage to the amplifier.

In an embodiment, the bias voltage applier may be turned off during the black period.

In an embodiment, the gamma reference voltage generator and the amplifier may be turned off during the black period.

In an embodiment of a display apparatus according to the present invention, the display apparatus includes: a display panel, a first channel, a first amplifier, A-1 switch, B-1 switch, a second channel, a second amplifier, A-2 switch and B-2 switch. The display panel includes a first display block and a second display block which extend in an extension direction of a data line, where the data line includes a first data line and a second data line. The first channel is connected to the first data line of the first display block. The first amplifier is configured to output a first data voltage to the first channel. The A-1 switch is connected between an output terminal of the first amplifier and the first channel. The B-1 switch is connected between an input terminal of a hold voltage and the first channel. The second channel is connected to the second data line of the second display block. The second amplifier is configured to output a second data voltage to the second channel. The A-2 switch is connected between an output terminal of the second amplifier and the second channel. The B-2 switch is connected between an input terminal of the hold voltage and the second channel. The A-1 switch and the B-1 switch are controlled independently from the A-2 switch and the B-2 switch.

In an embodiment, when the data line extends in a second direction, input image data have a second black pattern extending in the second direction and the second black pattern overlaps an entirety of the first display block, the A-1 switch may be turned off and the B-1 switch may be turned on.

In an embodiment, when the data line extends in a second direction, input image data have a second black pattern extending in the second direction and the second black pattern partially overlaps the first display block or does not overlap the first display block, the A-1 switch may be turned on and the B-1 switch may be turned off.

In an embodiment, when the data line extends in a second direction, input image data have a second black pattern extending in the second direction, the second black pattern overlaps an entirety of the first display block and the second black pattern partially overlaps the second display block or does not overlap the second display block, the A-1 switch may be turned off, the B-1 switch may be turned on, the A-2 switch may be turned on and the B-2 switch may be turned off.

In an embodiment, when the data line extends in a second direction, input image data have a third black pattern extending in a first direction perpendicular to the second direction in the first display block, the A-1 switch may be turned off and the B-1 switch may be turned on during a black period corresponding to the third black pattern and the A-1 switch may be turned on and the B-1 switch may be turned off during a normal period not corresponding to the third black pattern.

In an embodiment, a data signal may include a line configuration and line data. The line configuration may include a first display block setting signal representing the black period of the first display block and the normal period of the first display block and a second display block setting signal representing the black period of the second display block and the normal period of the second display block.

In an embodiment, the display apparatus may further include a gamma reference voltage generator configured to generate a gamma reference voltage, a digital-to-analog converter configured to convert a data signal to a data voltage based on the gamma reference voltage and a bias voltage applier configured to output a first bias voltage to the gamma reference voltage generator and to output a second bias voltage to the first amplifier and the second amplifier, where the data voltage includes the first data voltage and the second data voltage.

In an embodiment, while input image data have a black image for all display blocks in the display panel, the bias voltage applier may be turned off.

According to the display apparatus, when the display panel displays the black image, the bias voltage applier of the data driver may be turned off so that the constant current of the data driver may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

When the display panel displays a horizontal black pattern extending in the first direction, for a time corresponding to the horizontal black pattern, the outputs of the amplifiers may be blocked and the output of the data driver may be connected to a predetermined hold voltage so that the constant current of the data driver may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

When the display panel displays a vertical black pattern extending in the second direction, the outputs of the amplifiers corresponding to the vertical black pattern may be blocked and the output of the area of the data driver corresponding to the vertical black pattern may be connected to the predetermined hold voltage so that the constant current of the data driver may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

When the display block of the display panel displays a horizontal black pattern extending in the first direction, for a time corresponding to the horizontal black pattern, the outputs of the amplifiers corresponding to the display block may be blocked and the output of the data driver corresponding to the display block may be connected to the predetermined hold voltage so that the constant current of the data driver may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other features and advantages of the present invention will become more apparent by describing in detailed embodiments thereof with reference to the accompanying drawings, in which:

FIG. **1** is a block diagram illustrating a display apparatus according to an embodiment of the present invention;

FIG. **2** is a block diagram illustrating a driving controller, a data driver and a power voltage generator of FIG. **1**:

FIG. **3** is a diagram illustrating operations of the gamma reference voltage generator and the data driver of FIG. **1** in a normal mode:

FIG. **4** is a diagram illustrating operations of the gamma reference voltage generator and the data driver of FIG. **1** in a black mode:

FIG. **5** is a diagram illustrating a case in which a display panel of a display apparatus according to an embodiment of the present invention displays a first black pattern extending in a first direction;

FIG. **6** is a diagram illustrating a data signal transmitted from a driving controller of the display apparatus of FIG. **5** to a data driver of the display apparatus of FIG. **5**:

FIG. **7** is a diagram illustrating operations of a gamma reference voltage generator and a data driver corresponding to A**1** area of FIG. **5**;

FIG. **8** is a diagram illustrating operations of the gamma reference voltage generator and the data driver corresponding to A**2** area of FIG. **5**;

FIG. **9** is a diagram illustrating operations of the gamma reference voltage generator and the data driver corresponding to A**3** area of FIG. **5**;

FIG. **10** is a diagram illustrating a case in which a display panel of a display apparatus according to an embodiment of the present invention displays a second black pattern extending in a second direction;

FIG. **11** is a diagram illustrating operations of a gamma reference voltage generator and a data driver corresponding to the second black pattern of FIG. **10**;

FIG. **12** is a diagram illustrating a case in which the display panel of FIG. **10** is divided into eight display blocks:

FIG. **13** is a diagram illustrating a case in which the display panel of FIG. **10** is divided into eight display blocks and the display panel displays the second black pattern extending in the second direction:

FIG. **14** is a diagram illustrating a case in which a display block extending in a second direction in a display panel of a display apparatus according to an embodiment of the present invention displays a first black pattern extending in a first direction:

FIG. **15** is a diagram illustrating operations of a gamma reference voltage generator and a data driver corresponding to A**11** area, A**12** area and A**13** area of FIG. **14**;

FIG. **16** is a diagram illustrating operations of the gamma reference voltage generator and the data driver corresponding to A**21** area, A**22** area and A**23** area of FIG. **14**;

FIG. **17** is a diagram illustrating operations of the gamma reference voltage generator and the data driver corresponding to A**31** area, A**32** area and A**33** area of FIG. **14**;

FIG. **18** is a block diagram illustrating an electronic apparatus according to an embodiment of the present invention:

FIG. **19** is a diagram illustrating an example in which the electronic apparatus of FIG. **18** is implemented as a smart phone; and

FIG. **20** is a block diagram illustrating an electronic apparatus according to an embodiment of the present invention.

DETAILED DESCRIPTION

It will be understood that, although the terms "first," "second," "third" etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, "a first element," "component," "region," "layer" or "section" discussed below could be termed a second element, component, region, layer or section without departing from the teachings herein.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, "a", "an," "the," and "at least one" do not denote a limitation of quantity, and are intended to include both the singular and plural, unless the context clearly indicates otherwise. For example, "an element" has the same meaning as "at least one element," unless the context clearly indicates otherwise. "At least one" is not to be construed as limiting "a" or "an." "Or" means "and/or." As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms "comprises" and/or "comprising," or "includes" and/or "including" when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof. Hereinafter, the present invention will be explained in detail with reference to the accompanying drawings.

FIG. **1** is a block diagram illustrating a display apparatus according to an embodiment of the present invention.

Referring to FIG. **1**, the display apparatus includes a display panel **100** and a display panel driver. The display panel driver includes a driving controller **200**, a gate driver **300**, a gamma reference voltage generator **400** and a data driver **500**. The display panel driver may further include a power voltage generator **600**.

For example, the driving controller **200** and the data driver **500** may be integrally formed. In an embodiment, for example, the driving controller **200**, the gamma reference voltage generator **400** and the data driver **500** may be integrally formed. A driving module including at least the driving controller **200** and the data driver **500** which are integrally formed may be called to a timing controller embedded data driver ("TED").

The display panel **100** includes a display region AA on which an image is displayed and a peripheral region PA adjacent to the display region AA.

The display panel **100** includes a plurality of gate lines GL, a plurality of data lines DL and a plurality of pixels connected to the gate lines GL and the data lines DL. The gate lines GL may extend in a first direction D**1** and the data lines DL may extend in a second direction D**2** crossing the first direction D**1**.

The driving controller **200** receives input image data IMG and an input control signal CONT from an external apparatus. The input image data IMG may include red image data, green image data and blue image data. The input image data IMG may include white image data. The input image data IMG may include magenta image data, yellow image data and cyan image data. The input control signal CONT may include a master clock signal and a data enable signal. The input control signal CONT may further include a vertical synchronizing signal and a horizontal synchronizing signal.

The driving controller **200** generates a gate control signal CONT**1**, a data control signal CONT**2**, a gamma control signal CONT**3** and a data signal DATA based on the input image data IMG and the input control signal CONT.

The driving controller **200** generates the gate control signal CONT**1** for controlling an operation of the gate driver **300** based on the input control signal CONT, and outputs the gate control signal CONT**1** to the gate driver **300**. The gate control signal CONT**1** may further include a vertical start signal and a gate clock signal.

The driving controller **200** generates the data control signal CONT**2** for controlling an operation of the data driver **500** based on the input control signal CONT, and outputs the data control signal CONT**2** to the data driver **500**. The data control signal CONT**2** may include a horizontal start signal and a load signal.

The driving controller **200** generates the data signal DATA based on the input image data IMG. The driving controller **200** outputs the data signal DATA to the data driver **500**.

The driving controller **200** generates the gamma control signal CONT**3** for controlling an operation of the gamma reference voltage generator **400** based on the input control signal CONT, and outputs the gamma control signal CONT**3** to the gamma reference voltage generator **400**.

The gate driver **300** generates gate signals driving the gate lines GL in response to the gate control signal CONT**1** received from the driving controller **200**. The gate driver **300** outputs the gate signals to the gate lines GL. For example, the gate driver **300** may sequentially output the gate signals to the gate lines GL. In an embodiment, for example, the gate driver **300** may be mounted on the peripheral region PA of the display panel **100**. For example, the gate driver **300** may be integrated on the peripheral region PA of the display panel **100**.

The gamma reference voltage generator **400** generates a gamma reference voltage VGREF in response to the gamma control signal CONT**3** received from the driving controller **200**. The gamma reference voltage generator **400** provides the gamma reference voltage VGREF to the data driver **500**.

In an embodiment, the gamma reference voltage generator **400** may be disposed in the driving controller **200**, or in the data driver **500**.

The data driver **500** receives the data control signal CONT**2** and the data signal DATA from the driving controller **200**, and receives the gamma reference voltages VGREF from the gamma reference voltage generator **400**. The data driver **500** converts the data signal DATA into data voltages having an analog type using the gamma reference voltages VGREF. The data driver **500** outputs the data voltages to the data lines DL.

The power voltage generator **600** may generate a high display panel power voltage ELVDD and a low display panel power voltage ELVSS for driving the display panel **100** and output the high display panel power voltage ELVDD and the low display panel power voltage ELVSS to the display panel **100**. In addition, the power voltage generator **600** may generate a first driving voltage VLT for driving the driving controller **200** and output the first driving voltage VLT to the driving controller **200**. In addition, the power voltage generator **600** may generate a high data power voltage AVDD and a second driving voltage VLD for driving the data driver **500** and output the high data power voltage AVDD and the second driving voltage VLD to the data driver **500**.

FIG. **2** is a block diagram illustrating the driving controller **200**, the data driver **500** and the power voltage generator **600** of FIG. **1**.

Referring to FIGS. **1** and **2**, the data driver **500** may include a plurality of source driving chips SIC**1**, SIC**2**, SIC**3** and SIC**4**. Although the data driver **500** includes four source driving chips SIC**1**, SIC**2**, SIC**3** and SIC**4** in FIG. **2**, the present invention may not be limited thereto. Alternatively, the data driver **500** may include three or less source driving chips. Alternatively, the data driver **500** may include five or more source driving chips.

The driving controller **200** may output the data signal to the source driving chips SIC**1**, SIC**2**, SIC**3** and SIC**4** of the data driver **500** through an interface I/F.

The power voltage generator **600** may generate the first driving voltage VLT and output the first driving voltage VLT to the driving controller **200**. The power voltage generator **600** may generate the high data power voltage AVDD and the second driving voltage VLD and output the high data power voltage AVDD and the second driving voltage VLD to the data driver **500**.

FIG. **3** is a diagram illustrating operations of the gamma reference voltage generator **400** and the data driver **500** of FIG. **1** in a normal mode. FIG. **4** is a diagram illustrating operations of the gamma reference voltage generator **400** and the data driver **500** of FIG. **1** in a black mode.

Referring to FIGS. **1** to **4**, the data driver **500** may include a bias voltage applier **510**, a latch circuit **520**, a level shifter circuit **530**, a digital-to-analog converter circuit **540** and an amplifier circuit **550**.

The latch circuit **520** may include a plurality of latches LT**1** to LTN corresponding to data lines. The level shifter circuit **530** may include a plurality of level shifters LS**1** to LSN corresponding to data lines. The digital-to-analog converter circuit **540** may include a plurality of digital-to-analog converters DAC**1** to DACN corresponding to data lines. The amplifier circuit **550** may include a plurality of amplifiers AMP**1** to AMPN corresponding to data lines.

The latch circuit **520** may receive the data signal from the driving controller **200** and store the data signal.

The level shifter circuit **530** may convert the data signal having a low level to a high level data signal.

The digital-to-analog converter circuit **540** may convert the high level data signal to the data voltage having an analog type based on the gamma reference voltage VGREF.

The amplifier circuit **550** may output the data voltage to the data lines DL. The data lines DL may be connected to the amplifiers AMP**1** to AMPN through channels CH**1** to CHN.

The bias voltage applier **510** may output a first bias voltage VBG to the gamma reference voltage generator **400** and output a second bias voltage VBA to the amplifier circuit **550**.

In an embodiment, the gamma reference voltage generator **400** may be disposed in the data driver **500**. In an embodiment, the gamma reference voltage generator **400** may be disposed out of the data driver **500**.

Most of the constant current in the display panel driver is a current by the gamma reference voltage generator **400** and the amplifier circuit **550** operated by the high data power voltage AVDD.

The blocks **400**, **510**, **520**, **530**, **540** and **500** illustrated in FIG. **3** may operate to drive the display panel **100**. When the input image data IMG represent a black image, the amplifiers AMP**1** to AMPN may output a gamma voltage corresponding to a grayscale value of zero. Even in this case, the constant current of the gamma reference voltage generator **400** and the amplifier circuit **550** is consumed, and accordingly, the power consumption of the display apparatus may increase.

When the input image data IMG display only a black image on the display panel **100** for more than a specific time, some elements of the gamma reference voltage generator **400** and the data driver **500** may be turned off so that the constant current of the gamma reference voltage generator **400** and the data driver **500** may be reduced, and accordingly, the power consumption of the display apparatus may be effectively reduced.

In the present embodiment, during the input image data IMG represent the black image, the bias voltage applier **510** may be turned off. When the bias voltage applier **510** is turned off, the bias voltage applier **510** does not apply the bias voltages VBG and VBA to the gamma reference voltage generator **400** and the amplifier circuit **550** so that operations of the gamma reference voltage generator **400** and the amplifier circuit **550** may be floated. In an embodiment, for example, during the input image data IMG represent the black image, the gamma reference voltage generator **400** and the amplifier circuit **550** may be turned off.

Herein, whether the input image data IMG represent the black image may be determined on a frame-by-frame basis. When the input image data IMG have a black grayscale value for all pixels during a frame, the input image data IMG may represent the black image in the frame. Herein, the black grayscale value may not be limited to the grayscale value of zero. The black grayscale value may include a grayscale value less than a predetermined threshold grayscale value. When the input image data IMG have a grayscale value less than the predetermined threshold grayscale value for all pixels during a frame, the input image data IMG may represent the black image in the frame.

The display apparatus may further include the channels CH**1** to CHN, first switches SW**11** to SW**1**N connected between output terminals of the amplifiers AMP**1** to AMPN and the channels CH**1** to CHN and second switches SW**21** to SW**2**N connected between input terminal of a hold voltage HV and the channels CH**1** to CHN.

Herein, the hold voltage HV may be the high data power voltage AVDD generated by the power voltage generator **600**. However, the hold voltage may not be limited to the high data power voltage AVDD generated by the power voltage generator **600**. In an embodiment, for example, the hold voltage may be a low data power voltage AVSS in a certain mode of the display panel **100**. The hold voltage may be a predetermined direct current ("DC") voltage.

FIG. **3** represents a normal mode in which the input image data IMG includes image which is not the black image. In the normal mode, the input image data IMG may include a grayscale value which is not the black grayscale value.

As shown in FIG. **3**, in the normal mode, the first switches SW**11** to SW**1**N may be turned on and the second switches SW**21** to SW**2**N may be turned off. In this case, the data lines DL are connected to the amplifiers AMP**1** to AMPN through the channels CH**1** to CHN and the first switches SW**11** to SW**1**N and grayscale value voltages corresponding to the data lines DL may be applied to the data lines DL.

FIG. **4** represents a black mode in which the input image data IMG only includes the black image.

As shown in FIG. **4**, in the black mode, the first switches SW**11** to SW**1**N may be turned off and the second switches SW**21** to SW**2**N may be turned on. In this case, the data lines DL are connected to the input terminal of the hold voltage HV through the channels CH**1** to CHN and the second switches SW**21** to SW**2**N and the hold voltage HV may be commonly applied to the data lines DL.

According to the present embodiment, when the display panel **100** displays the black image, the bias voltage applier **510** of the data driver **500** may be turned off so that the constant current of the data driver **500** may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

FIG. **5** is a diagram illustrating a case in which a display panel **100** of a display apparatus according to an embodiment of the present invention displays a first black pattern extending in a first direction. FIG. **6** is a diagram illustrating a data signal transmitted from a driving controller **200** of the display apparatus of FIG. **5** to a data driver **500** of the display apparatus of FIG. **5**. FIG. **7** is a diagram illustrating operations of a gamma reference voltage generator **400** and the data driver **500** corresponding to A**1** area of FIG. **5**. FIG. **8** is a diagram illustrating operations of the gamma reference voltage generator **400** and the data driver **500** corresponding to A**2** area of FIG. **5**. FIG. **9** is a diagram illustrating operations of the gamma reference voltage generator **400** and the data driver **500** corresponding to A**3** area of FIG. **5**.

The display apparatus according to the present embodiment are substantially the same as the display apparatus of the previous embodiment explained referring to FIGS. **1** to **4** except that the hold voltage is applied to the data lines during a black period corresponding to a horizontal black pattern when the horizontal black pattern (a first black pattern) is displayed. Thus, the same reference numerals will be used to refer to the same or like parts as those described in the previous embodiment of FIGS. **1** to **4** and any repetitive explanation concerning the above elements will be omitted.

In FIG. **5**, a white image (a normal image) may be displayed on a first area (A**1** area) of the display panel **100**, a black image may be displayed on a second area (A**2** area) of the display panel **100** and a white image (a normal image) may be displayed on a third area (A**3** area) of the display panel **100**. Herein, the first area (A**1** area), the second area (A**2** area) and the third area (A**3** area) may extend in a direction (the first direction D**1**) perpendicular to an extension direction of the data line DL. The first black pattern displayed on the second area (A**2** area) may extend in the first direction D**1**. For convenience of explanation, the first black pattern may be referred to as a horizontal black pattern. Herein, the normal image may mean an image which is not the black image. When the input image data IMG have at least one grayscale value equal to or greater than the predetermined threshold grayscale value during a frame, the input image data IMG may represent the normal image in the frame.

Referring to FIGS. **1** to **9**, the display apparatus may include a display panel **100** including data lines DL, channels CH**1** to CHN connected to the data lines DL and amplifiers AMP**1** to AMPN for outputting data voltages, first switches SW**11** to SW**1**N connected between output terminals of the amplifiers AMP**1** to AMPN and the channels CH**1** to CHN and second switches SW**21** to SW**2**N connected between input terminal of a hold voltage HV and the channels CH**1** to CHN.

When the data lines DL extend in the second direction D**2** and the input image data IMG have a first black pattern extending in the first direction D**1** perpendicular to the second direction D**2**, the first switches SW**11** to SW**1**N may be turned off and the second switches SW**21** to SW**2**N may be turned on during a black period corresponding to the first black pattern and the first switches SW**11** to SW**1**N may be turned on and the second switches SW**21** to SW**2**N may be turned off during a normal period not corresponding to the first black pattern.

FIG. **7** may correspond to the first area (A**1** area) and the first area (A**1** area) may correspond to the normal period in which the input image data IMG have the white image (the normal image).

As shown in FIG. **7**, in the normal period, the first switches SW**11** to SW**1**N may be turned on and the second switches SW**21** to SW**2**N may be turned off. In this case, the data lines DL are connected to the amplifiers AMP**1** to AMPN through the channels CH**1** to CHN and the first switches SW**11** to SW**1**N and grayscale value voltages corresponding to the data lines DL may be applied to the data lines DL.

FIG. **8** may correspond to the second area (A**2** area) and the second area (A**2** area) may correspond to the black period in which the input image data IMG have only the black image.

As shown in FIG. **8**, in the black period, the first switches SW**11** to SW**1**N may be turned off and the second switches SW**21** to SW**2**N may be turned on. In this case, the data lines DL are connected to the input terminal of the hold voltage HV through the channels CH**1** to CHN and the second switches SW**21** to SW**2**N and the hold voltage HV may be commonly applied to the data lines.

FIG. **9** may correspond to the third area (A**3** area) and the third area (A**3** area) may correspond to the normal period in which the input image data IMG have the white image (the normal image).

As shown in FIG. **9**, in the normal period, the first switches SW**11** to SW**1**N may be turned on and the second switches SW**21** to SW**2**N may be turned off. In this case, the data lines DL are connected to the amplifiers AMP**1** to AMPN through the channels CH**1** to CHN and the first switches SW**11** to SW**1**N and grayscale value voltages corresponding to the data lines DL may be applied to the data lines DL.

As shown in FIG. **6**, the data signal DATA may include a line configuration LC and line data LDATA. A setting signal representing the black period and the normal period may be included in the line configuration LC. The setting signal may be transmitted from the driving controller **200** to the data driver **500** in the line configuration LC format.

In addition, when proceeding from the black period to the normal period, it may be hard that the first switches SW**11** to SW**1**N and the second switches SW**21** to SW**2**N, the bias voltage applier **510**, the gamma reference voltage generator

400 and the amplifier circuit **550** are turned on and operate a normal operation within a short period of time. Thus, when proceeding from the black period to the normal period, the setting signal may be determined based on a wakeup time. When proceeding from the black period to the normal period, the setting signal may indicate the normal period corresponding to a horizontal line several horizontal lines before a transition horizontal line.

In the present embodiment, in the black period, the bias voltage applier **510** may be turned off. In addition, in the black period, the gamma reference voltage generator **400** and the amplifiers AMP**1** to AMPN may be turned off.

According to the present embodiment, when the display panel **100** displays the black image, the bias voltage applier **510** of the data driver **500** may be turned off so that the constant current of the data driver **500** may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

When the display panel **100** displays the horizontal black pattern extending in the first direction D**1**, for a time corresponding to the horizontal black pattern, the outputs of the amplifiers AMP**1** to AMPN may be blocked and the output of the data driver **500** may be connected to the predetermined hold voltage HV so that the constant current of the data driver **500** may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

FIG. **10** is a diagram illustrating a case in which a display panel **100** of a display apparatus according to an embodiment of the present invention displays a second black pattern extending in a second direction D**2**. FIG. **11** is a diagram illustrating operations of a gamma reference voltage generator **400** and a data driver **500** corresponding to the second black pattern of FIG. **10**. FIG. **12** is a diagram illustrating a case in which the display panel **100** of FIG. **10** is divided into eight display blocks. FIG. **13** is a diagram illustrating a case in which the display panel **100** of FIG. **10** is divided into eight display blocks and the display panel **100** displays the second black pattern extending in the second direction D**2**.

The display apparatus according to the present embodiment are substantially the same as the display apparatus of the previous embodiment explained referring to FIGS. **1** to **4** except that the hold voltage is applied to the data lines corresponding to a vertical black pattern when the vertical black pattern (a second black pattern) is displayed. Thus, the same reference numerals will be used to refer to the same or like parts as those described in the previous embodiment of FIGS. **1** to **4** and any repetitive explanation concerning the above elements will be omitted.

In FIG. **10**, a white image (a normal image) may be displayed on a first display block (B**1** area) of the display panel **100**, a black image may be displayed on a second display block (B**2** area) of the display panel **100** and a white image (a normal image) may be displayed on a third display block (B**3** area) of the display panel **100**. Herein, the first display block (B**1** area), the second display block (B**2** area) and the third display block (B**3** area) may extend in an extension direction (the second direction D**2**) of the data line DL. The second black pattern displayed on the second display block (B**2** area) may extend in the second direction D**2**. For convenience of explanation, the second black pattern may be referred to as a vertical black pattern.

Referring to FIGS. **1** to **4** and **10** to **13**, the display apparatus may include a display panel **100** including the first display block (B**1** area) and the second display block (B**2** area), a first channel CH**11** and CH**12** connected to the data line DL in the first display block (B**1** area), a first amplifier AMP**11** and AMP**12** for outputting a data voltage to the first channel CH**11** and CH**12**, A-1 switch SW**111** and SW**112** connected between an output terminal of the first amplifier AMP**11** and AMP**12** and the first channel CH**11** and CH**12**, B-1 switch SW**211** and SW**212** connected between an input terminal of the hold voltage HV and the first channel CH**11** and CH**12**, a second channel CH**21** and CH**22** connected to the data line DL in the second display block (B**2** area), a second amplifier AMP**21** and AMP**22** for outputting a data voltage to the second channel CH**21** and CH**22**, A-2 switch SW**121** and SW**122** connected between an output terminal of the second amplifier AMP**21** and AMP**22** and the second channel CH**21** and CH**22** and B-2 switch SW**221** and SW**222** connected between an input terminal of the hold voltage HV and the second channel CH**21** and CH**22**.

In the present embodiment, the A-1 switch SW**11*l*** and SW**112** and the B-1 switch SW**211** and SW**212** may be controlled independently from the A-2 switch SW**121** and SW**122** and the B-2 switch SW**221** and SW**222**.

The A-1 switch SW**111** and SW**112** and the B-1 switch SW**211** and SW**212** may be controlled dependently. In an embodiment, for example, when the A-1 switch SW**111** and SW**112** is turned on, the B-1 switch SW**211** and SW**212** may be turned off. When the A-1 switch SW**111** and SW**112** is turned off, the B-1 switch SW**211** and SW**212** may be turned on.

Similarly, the A-2 switch SW**121** and SW**122** and the B-2 switch SW**221** and SW**222** may be controlled dependently.

In addition, the display panel **100** may further include a third display block (B**3** area) and the display apparatus may further include channels CH**31** and CH**32**, amplifiers AMP**31** and AMP**32** and switches SW**131**, SW**132**, SW**231** and SW**232** corresponding the third display block (B**3** area).

In this case, the A-1 switch SW**111** and SW**112** and the B-1 switch SW**211** and SW**212** may be controlled independently from the A-2 switch SW**121** and SW**122** and the B-2 switch SW**221** and SW**222**, the A-1 switch SW**111** and SW**112** and the B-1 switch SW**211** and SW**212** may be controlled independently from the A-3 switch SW**131** and SW**132** and the B-3 switch SW**231** and SW**232**, and the A-2 switch SW**121** and SW**122** and the B-2 switch SW**221** and SW**222** may be controlled independently from the A-3 switch SW**131** and SW**132** and the B-3 switch SW**231** and SW**232**.

In an embodiment, for example, when the data line DL extends in the second direction D**2**, the input image data IMG have a second black pattern extending in the second direction D**2** and the second black pattern overlaps an entirety of the first display block (B**1** area), the A-1 switch SW**111** and SW**112** corresponding to the first display block may be turned off and the B-1 switch SW**211** and SW**212** corresponding to the first display block may be turned on. The channels CH**11** and CH**12** corresponding to the first display block (B**1** area) may output the hold voltage HV.

In an embodiment, for example, when the data line DL extends in the second direction D**2**, the input image data IMG have a second black pattern extending in the second direction D**2** and the second black pattern overlaps an entirety of the second display block (B**2** area), the A-2 switch SW**121** and SW**122** corresponding to the second display block may be turned off and the B-2 switch SW**221** and SW**222** corresponding to the second display block may be turned on. The channels CH**21** and CH**22** corresponding to the second display block (B**2** area) may output the hold voltage HV.

In an embodiment, for example, when the data line DL extends in the second direction D2, the input image data IMG have a second black pattern extending in the second direction D2 and the second black pattern partially overlaps the first display block (B1 area) or does not overlap the first display block (B1 area), the A-1 switch SW11*l* and SW112 corresponding to the first display block may be turned on and the B-1 switch SW211 and SW212 corresponding to the first display block may be turned off. The channels CH11 and CH12 corresponding to the first display block may output the normal data voltages.

In an embodiment, for example, when the data line DL extends in the second direction D2, the input image data IMG have a second black pattern extending in the second direction D2 and the second black pattern partially overlaps the second display block (B2 area) or does not overlap the second display block (B2 area), the A-2 switch SW121 and SW122 corresponding to the second display block may be turned on and the B-2 switch SW221 and SW222 corresponding to the second display block may be turned off. The channels CH21 and CH22 corresponding to the second display block may output the normal data voltages.

The switches independently operate for the display blocks so that some switches may operate to output the normal data voltages and some switches may output the hold voltage HV.

When the data line DL extends in the second direction D2, the input image data IMG have a second black pattern extending in the second direction D2, the second black pattern overlaps the entirety of the first display block, and the second black pattern partially overlaps the second display block or does not overlap the second display block, the A-1 switch SW111 and SW112 may be turned off, the B-1 switch SW211 and SW212 may be turned on, the A-2 switch SW121 and SW122 may be turned on and the B-2 switch SW221 and SW222 may be turned off. The channels CH11 and CH12 corresponding to the first display block may output the hold voltage and the channels CH21 and CH22 corresponding to the second display block may output the normal data voltage.

In FIG. 12, the display panel 100 may be divided into eight display blocks BL1 to BL8. Herein, the data driver 500 may be divided into data output blocks.

When the vertical black pattern is displayed like FIG. 13, data output blocks corresponding to a fourth display block BL4 and a fifth display block BL5 may output the hold voltage. In contrast, data output blocks corresponding to display blocks except for the fourth display block BL4 and the fifth display block BL5 may output the normal data voltages.

In the present embodiment, while the input image data IMG have the black image for all display blocks in the display panel 100, the bias voltage applier 510 may be turned off.

In addition, while the input image data IMG have the black image for all display blocks in the display panel 100, the gamma reference voltage generator 400 and the amplifier circuit 550 may also be turned off.

According to the present embodiment, when the display panel 100 displays the black image, the bias voltage applier 510 of the data driver 500 may be turned off so that the constant current of the data driver 500 may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

When the display panel 100 displays the vertical black pattern extending in the second direction D2, the outputs of the amplifiers corresponding to the vertical black pattern may be blocked and the output of the area of the data driver 500 corresponding to the vertical black pattern may be connected to the predetermined hold voltage HV so that the constant current of the data driver 500 may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

FIG. 14 is a diagram illustrating a case in which a display block extending in a second direction D2 in a display panel 100 of a display apparatus according to an embodiment of the present invention displays a first black pattern extending in a first direction D1. FIG. 15 is a diagram illustrating operations of a gamma reference voltage generator 400 and a data driver 500 corresponding to A11 area, A12 area and A13 area of FIG. 14. FIG. 16 is a diagram illustrating operations of the gamma reference voltage generator 400 and the data driver 500 corresponding to A21 area, A22 area and A23 area of FIG. 14. FIG. 17 is a diagram illustrating operations of the gamma reference voltage generator 400 and the data driver 500 corresponding to A31 area, A32 area and A33 area of FIG. 14.

The display apparatus according to the present embodiment are substantially the same as the display apparatus of the previous embodiment explained referring to FIGS. 1 to 4 except that an embodiment of applying the hold voltage to the data lines during a black period corresponding to a horizontal black pattern when the horizontal black pattern (a first black pattern) is displayed and an embodiment of applying the hold voltage to the data lines corresponding to a vertical black pattern when the vertical black pattern (a second black pattern) is displayed are combined. Thus, the same reference numerals will be used to refer to the same or like parts as those described in the previous embodiment of FIGS. 1 to 4 and any repetitive explanation concerning the above elements will be omitted.

In FIG. 14, a black image may be displayed on a fifth display area (A22 area) of the display panel 100, and normal image may be displayed on display areas (A11, A12, A13, A21, A23, A31, A32 and A33 areas) except for the fifth display area (A22 area).

Referring to FIGS. 1 to 15, in the present embodiment, the A-1 switch SW111 and SW112 and the B-1 switch SW211 and SW212 corresponding to the A11, A21 and A31 areas may be controlled independently from the A-2 switch SW121 and SW122 and the B-2 switch SW221 and SW222 corresponding to the A12, A22 and A32 areas.

In an embodiment, for example, when the data line DL extends in the second direction D2, the input image data IMG have a third black pattern extending in the first direction D1 perpendicular to the second direction D2 in the second display block A12, A22 and A32 areas, the A-2 switch may be turned off and the B-2 switch may be turned on during a black period corresponding to the third black pattern and the A-2 switch may be turned on and the B-2 switch may be turned off during a normal period not corresponding to the third black pattern.

The normal period corresponding to A11, A12 and A13 areas of FIG. 14 may be substantially the same as the normal period of FIG. 7. Thus, in the normal period, the first switches SW11 to SW1N may be turned on and the second switches SW21 to SW2N may be turned off during the normal period. In this case, the data lines DL are connected to the amplifiers AMP1 to AMPN through the channels CH1 to CHN and the first switches SW11 to SW1N and grayscale value voltages corresponding to the data lines DL may be applied to the data lines DL.

The normal period corresponding to A31, A32 and A33 areas of FIG. 14 may be substantially the same as the normal period of FIG. 9. Thus, in the normal period, the first switches SW**11** to SW**1**N may be turned on and the second switches SW**21** to SW**2**N may be turned off during the normal period. In this case, the data lines DL are connected to the amplifiers AMP**1** to AMPN through the channels CH**1** to CHN and the first switches SW**11** to SW**1**N and grayscale value voltages corresponding to the data lines DL may be applied to the data lines DL.

A partial black period corresponding to A**21**, A**22** and A**23** areas of FIG. **14** may be substantially the same as the period of FIG. **11**. Thus, the A-1 switch SW**111** and SW**112** and the B-1 switch SW**211** and SW**212** may be controlled independently from the A-2 switch SW**121** and SW**122** and the B-2 switch SW**221** and SW**222**, the A-1 switch SW**111** and SW**112** and the B-1 switch SW**211** and SW**212** may be controlled independently from the A-3 switch SW**131** and SW**132** and the B-3 switch SW**231** and SW**232**, and the A-2 switch SW**121** and SW**122** and the B-2 switch SW**221** and SW**222** may be controlled independently from the A-3 switch SW**131** and SW**132** and the B-3 switch SW**231** and SW**232**.

When only the A**22** area includes the black image as shown in FIG. **14**, the channels CH**21** and CH**22** corresponding to the A**22** area may output the hold voltage, the channels CH**11** and CH**12** corresponding to the A**21** area and the channels CH**31** and CH**32** corresponding to the A**23** area may output the normal data voltage for a time period corresponding to the A**21**, A**22** and A**23** areas.

As shown in FIG. **6**, the data signal may include a line configuration LC and line data LDATA. A setting signal representing the black period and the normal period may be included in the line configuration LC. The setting signal may be transmitted from the driving controller **200** to the data driver **500** in the line configuration LC format.

In the present embodiment, the black period and the normal period is set for each line and the black output and the normal output may be set for each output block of the data driver **500** according to the display blocks of the display panel **100** in the second direction D**2**.

Thus, in the present embodiment, the line configuration may include information of the display block. When the display panel **100** includes a first display block and a second display block extending in the second direction D**2**, the line configuration may include a first display block setting signal representing the black period of the first display block and the normal period of the first display block and a second display block setting signal representing the black period of the second display block and the normal period of the second display block.

According to the present embodiment, when the display panel **100** displays the black image, the bias voltage applier **510** of the data driver **500** may be turned off so that the constant current of the data driver **500** may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

When the display panel **100** displays the horizontal black pattern extending in the first direction D**1**, for a time corresponding to the horizontal black pattern, the outputs of the amplifiers AMP**1** to AMPN may be blocked and the output of the data driver **500** may be connected to the predetermined hold voltage HV so that the constant current of the data driver **500** may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

When the display panel **100** displays the vertical black pattern extending in the second direction D**2**, the outputs of the amplifiers corresponding to the vertical black pattern may be blocked and the output of the area of the data driver **500** corresponding to the vertical black pattern may be connected to the predetermined hold voltage HV so that the constant current of the data driver **500** may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

When the display block of the display panel **100** extending in the second direction D**2** displays the horizontal black pattern extending in the first direction D**1**, for a time corresponding to the horizontal black pattern, the outputs of the amplifiers corresponding to the display block may be blocked and the output of the area of the data driver **500** corresponding to the display block may be connected to the predetermined hold voltage HV so that the constant current of the data driver **500** may be reduced, and accordingly the power consumption of the display apparatus may be effectively reduced.

FIG. **18** is a block diagram illustrating an electronic apparatus according to an embodiment of the present invention. FIG. **19** is a diagram illustrating an example in which the electronic apparatus of FIG. **18** is implemented as a smart phone.

Referring to FIGS. **18** and **19**, the electronic apparatus **1000** may include a processor **1010**, a memory device **1020**, a storage device **1030**, an input/output (I/O) device **1040**, a power supply **1050**, and a display apparatus **1060**. Here, the display apparatus **1060** may be the display apparatus of FIG. **1**. In addition, the electronic apparatus **1000** may further include a plurality of ports for communicating with a video card, a sound card, a memory card, a universal serial bus ("USB") device, other electronic apparatuses, etc.

In an embodiment, as illustrated in FIG. **19**, the electronic apparatus **1000** may be implemented as a smart phone. However, the electronic apparatus **1000** is not limited thereto. For example, the electronic apparatus **1000** may be implemented as a cellular phone, a video phone, a smart pad, a smart watch, a tablet PC, a car navigation system, a computer monitor, a laptop, a head mounted display ("HMD") device, or the like.

The processor **1010** may perform various computing functions or various tasks. The processor **1010** may be a micro-processor, a central processing unit ("CPU"), an application processor ("AP"), or the like. The processor **1010** may be coupled to other components via an address bus, a control bus, a data bus, etc. Further, the processor **1010** may be coupled to an extended bus such as a peripheral component interconnection ("PCI") bus.

The processor **1010** may output the input image data IMG and the input control signal CONT to the driving controller **200** of FIG. **1**.

The memory device **1020** may store data for operations of the electronic apparatus **1000**. For example, the memory device **1020** may include at least one non-volatile memory device such as an erasable programmable read-only memory ("EPROM") device, an electrically erasable programmable read-only memory ("EEPROM") device, a flash memory device, a phase change random access memory ("PRAM") device, a resistance random access memory ("RRAM") device, a nano floating gate memory ("NFGM") device, a polymer random access memory ("PoRAM") device, a magnetic random access memory ("MRAM") device, a ferroelectric random access memory ("FRAM") device, and the like and/or at least one volatile memory device such as a dynamic random access memory ("DRAM") device, a static random access memory ("SRAM") device, a mobile DRAM device, and the like.

The storage device **1030** may include a solid state drive ("SSD") device, a hard disk drive ("HDD") device, a CD-

ROM device, or the like. The I/O device **1040** may include an input device such as a keyboard, a keypad, a mouse device, a touch-pad, a touch-screen, and the like and an output device such as a printer, a speaker, and the like. In some embodiments, the display apparatus **1060** may be included in the I/O device **1040**. The power supply **1050** may provide power for operations of the electronic apparatus **1000**. The display apparatus **1060** may be coupled to other components via the buses or other communication links.

FIG. **20** is a block diagram illustrating an electronic apparatus **101** according to an embodiment of the present invention.

Referring to FIGS. **1** to **20**, an electronic apparatus **101** outputs various information through a display module **140** in an operating system. When a processor **110** executes an application stored in a memory **120**, the display module **140** provides application information to a user through a display panel **141**.

The processor **110** obtains an external input through an input module **130** or a sensor module **161** and executes an application corresponding to the external input. In an embodiment, for example, when the user selects a camera icon displayed on the display panel **141**, the processor **110** obtains a user input through an input sensor **161-2** and activates a camera module **171**. The processor **110** transfers image data corresponding to a captured image obtained through the camera module **171** to the display module **140**. The display module **140** may display an image corresponding to the captured image through the display panel **141**.

In an embodiment, when a personal information authentication is executed in the display module **140**, a fingerprint sensor **161-1** obtains input fingerprint information as input data. The processor **110** compares input data obtained through the fingerprint sensor **161-1** with authentication data stored in the memory **120**, and executes an application according to a comparison result. The display module **140** may display information executed according to application logic through the display panel **141**.

In an embodiment, when a music streaming icon displayed on the display module **140** is selected, the processor **110** obtains a user input through the input sensor **161-2** and activates a music streaming application stored in the memory **120**. When a music execution command is input in the music streaming application, the processor **110** activates a sound output module **163** to provide sound information corresponding to the music execution command to the user.

In the above, the operation of the electronic apparatus **101** is briefly described. Hereinafter, a configuration of the electronic apparatus **101** is described in detail. Some of elements of the electronic apparatus **101** described later may be integrated and provided as one element, or one element may be separated as two or more elements.

The electronic apparatus **101** may communicate with an external electronic apparatus **102** through a network (e.g., a short-range wireless communication network or a long-range wireless communication network). According to an embodiment, the electronic apparatus **101** may include the processor **110**, the memory **120**, the input module **130**, the display module **140**, a power module **150**, an embedded module **160**, and an external module **170**. According to an embodiment, in the electronic apparatus **101**, at least one of the above-described elements may be omitted or one or more other apparatus may be added.

According to an embodiment, some of the above-described elements (e.g., the sensor module **161**, an antenna module **162** or the sound output module **163**) may be integrated into another element (e.g., the display module **140**).

The processor **110** may execute software to control at least one other element (e.g., hardware or software element) of the electronic apparatus **101** connected to the processor **110** and to perform various data processing or operations. According to an embodiment, as at least part of the data processing or the operations, the processor **110** may store receive instructions or data from other elements (e.g. the input module **130**, the sensor module **161** or a communication module **173**) in a volatile memory **121**, may process the instructions or data stored in the volatile memory **121** and may store result data of the processing in a nonvolatile memory **122**.

The processor **110** may include a main processor **111** and an auxiliary processor **112**. The main processor **111** may include at least one of a central processing unit (CPU) **111-1** and an application processor (AP). The main processor **111** may further include any one or more of a graphic processing unit ("GPU") **111-2**, a communication processor ("CP") and an image signal processor ("ISP"). The main processor **111** may further include a neural processing unit ("NPU") **111-3**. The neural network processing unit **111-3** is a processor specialized in processing an artificial intelligence model. The artificial intelligence model may be generated through a machine learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be one of a deep neural network ("DNN"), a convolutional neural network ("CNN"), a recurrent neural network ("RNN"), a restricted boltzmann machine ("RBM"), a deep belief network ("DBN"), a bidirectional recurrent deep neural network ("BRDNN") and a deep Q-networks or a combination of two or more of the above. However, the artificial neural network is not limited to the above examples. The artificial intelligence model may include software structures, in addition to hardware structures or instead of the hardware structures. At least two of the above-described processing units and the above-described processors may be implemented as an integrated element (e.g., a single chip) or each may be implemented as independent elements (e.g., in a plurality of chips).

The auxiliary processor **112** may include a controller. The controller may include an interface conversion circuit and a timing control circuit. The controller receives an image signal from the main processor **111**, converts a data format of the image signal to meet interface specifications with the display module **140**, and outputs image data. The controller may output various control signals for driving the display module **140**.

The auxiliary processor **112** may further include a data converting circuit **112-2**, a gamma correction circuit **112-3** and a rendering circuit **112-4**. The data converting circuit **112-2** may receive the image data from the controller and may compensate the image data such that the image is displayed with a desired luminance according to characteristics of the electronic apparatus **101** or a user setting or may convert the image data to reduce a power consumption or compensate for afterimages. The gamma correction circuit **112-3** may convert the image data or a gamma reference voltage such that the image displayed on the electronic apparatus **101** has desired gamma characteristics. The rendering circuit **112-4** may receive the image data from the controller and may render the image data based on a pixel arrangement of the display panel **141** included in the electronic apparatus **101**. At least one of the data converting circuit **112-2**, the gamma correction circuit **112-3** and the rendering circuit **112-4** may be integrated into another element (e.g., the main processor **111** or the controller). At least one of the data converting circuit **112-2**, the gamma correction circuit **112-3** and the rendering circuit **112-4** may be integrated into a data driver **143** to be described later.

The memory **120** may store various data used by at least one element (e.g., the processor **110** or the sensor module **161**) of the electronic apparatus **101** and input data or output data for commands related thereto. The memory **120** may include at least one of the volatile memory **121** and the nonvolatile memory **122**.

The input module **130** may receive commands or data used to the elements (e.g., the processor **110**, the sensor module **161** or the sound output module **163**) of the electronic apparatus **101** from the outside of the electronic apparatus **101** (e.g., the user or the external electronic apparatus **102**).

The input module **130** may include a first input module **131** for receiving commands or data from the user and a second input module **132** for receiving commands or data from the external electronic apparatus **102**. The first input module **131** may include a microphone, a mouse, a keyboard, a key (e.g., a button) or a pen (e.g., a passive pen or an active pen). The second input module **132** may support a designated protocol capable of connecting to the external electronic apparatus **102** by wire or wirelessly. According to an embodiment, the second input module **132** may include a high definition multimedia interface ("HDMI"), a universal serial bus ("USB") interface, an SD card interface or an audio interface. The second input module **132** may include a connector physically connected to the external electronic apparatus **102**, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

The display module **140** visually provides information to the user. The display module **140** may include the display panel **141**, a scan driver **142** and the data driver **143**. The display module **140** may further include a window, a chassis and a bracket to protect the display panel **141**.

The display panel **141** may include a liquid crystal display panel, an organic light emitting display panel or an inorganic light emitting display panel. A type of the display panel **141** is not particularly limited. The display panel **141** may be a rigid type or a flexible type capable of being rolled or folded. The display module **140** may further include a supporter or a heat dissipation member supporting the display panel **141**.

The scan driver **142** may be mounted on the display panel **141** as a driving chip. Alternatively, the scan driver **142** may be integrated on the display panel **141**. In an embodiment, for example, the scan driver **142** may include an amorphous silicon TFT gate driver circuit ("ASG") integrated on the display panel **141**, a low temperature polycrystalline silicon ("LTPS") TFT gate driver circuit integrated on the display panel **141**, or an oxide semiconductor TFT gate driver circuit ("OSG") integrated on the display panel **141**. The scan driver **142** receives a control signal from the controller and outputs the scan signals to the display panel **141** in response to the control signal.

The display module **140** may further include a light emission driver. The light emission driver outputs a light emission control signal to the display panel **141** in response to a control signal received from the controller. The light emission driver may be formed independently from the scan driver **142**. Alternatively, the light emission driver and the scan driver **142** may be integrally formed.

The data driver **143** receives a control signal from the controller and converts the image data into an analog voltage (e.g., the data voltage) and output the data voltages to the display panel **141** in response to the control signal.

The data driver **143** may be integrated into another element (e.g., the controller). The functions of the interface conversion circuit and the timing control circuit of the controller described above may be integrated into the data driver **143**.

The display module **140** may further include a voltage generating circuit. The voltage generating circuit may output various voltages for driving the display panel **141**.

The power module **150** supplies power to elements of the electronic apparatus **101**. The power module **150** may include a battery which supplies a power voltage. The battery may include a non-rechargeable primary cell, a rechargeable secondary cell or a fuel cell. The power module **150** may include a power management integrated circuit ("PMIC"). The PMIC supplies optimized power to each of the above-described modules and modules described later. The power module **150** may include a wireless power transmission/reception member electrically connected to the battery. The wireless power transmission/reception member may include a plurality of antenna radiators in a form of coils.

The electronic apparatus **101** may further include the embedded module **160** and the external module **170**. The embedded module **160** may include the sensor module **161**, the antenna module **162** and the sound output module **163**. The external module **170** may include the camera module **171**, a light module **172** and the communication module **173**.

The sensor module **161** may detect an input by a user's body or an input by the pen among the first input module **131**, and generate an electrical signal or data value corresponding to the input. The sensor module **161** may include at least one of the fingerprint sensor **161-1**, the input sensor **161-2** and a digitizer **161-3**.

The fingerprint sensor **161-1** may generate a data value corresponding to a user's fingerprint. The fingerprint sensor **161-1** may include one of an optical fingerprint sensor or a capacitive fingerprint sensor.

The input sensor **161-2** may generate data values corresponding to coordinate information of the input by the user's body or the input by the pen. The input sensor **161-2** generates a capacitance change due to an input as a data value. The input sensor **161-2** may detect an input by the passive pen or transmit/receive data to/from the active pen.

The input sensor **161-2** may measure bio signals such as a blood pressure, a moisture, or a body fat. For example, when a user touches a part of his body to a sensor layer or a sensing panel and does not move for a certain period of time, the input sensor **161-2** may detect the bio signal based on a change in an electric field caused by the part of the body so that the display module **140** may output user's desired information.

The digitizer **161-3** may generate a data value corresponding to the coordinate information input by the pen. The digitizer **161-3** generates an amount of electromagnetic change by the input as a data value. The digitizer **161-3** may detect an input by the passive pen or transmit/receive data to/from the active pen.

At least one of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** may be formed as a sensor layer on the display panel **141** through a continuous process. The fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** may be disposed on the display panel **141**. At least one of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3**, for example, the digitizer **161-3**, may be disposed under the display panel **141**.

At least two or more of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** may be integrated into the sensing panel through the same process. When at least two or more of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** are integrated into the sensing panel, the sensing panel may be disposed between the display panel **141** and a window disposed over an upper surface of the display panel **141**. According to an embodiment, the sensing panel may be disposed on the window. The present invention may not be limited to a position of the sensing panel.

At least one of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** may be embedded in the display panel **141**. In an embodiment, for example, at least one of the fingerprint sensor **161-1**, the input sensor **161-2** and the digitizer **161-3** is formed simultaneously with the display panel **141** through a process of forming elements included in the display panel **141** (e.g., light emitting elements, transistors, etc.).

In addition, the sensor module **161** may generate an electrical signal or a data value corresponding to an internal state or an external state of the electronic apparatus **101**. In an embodiment, for example, the sensor module **161** may further include a gesture sensor, a gyro sensor, a barometric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an IR (infrared) sensor, a biosensor, a temperature sensor, a humidity sensor or an illuminance sensor.

The antenna module **162** may include one or more antennas for transmitting a signal or power to outside or receiving a signal or power from outside. According to an embodiment, the communication module **173** may transmit a signal to an external electronic apparatus or receive a signal from an external electronic apparatus through an antenna suitable for a communication method. An antenna pattern of the antenna module **162** may be integrated with an element of the display module **140** (e.g., the display panel **141**) or the input sensor **161-2**.

The sound output module **163** is a device for outputting sound signals to the outside of the electronic apparatus **101**. In an embodiment, for example, the sound output module **163** may include a speaker used for general purposes such as playing multimedia or recording and a receiver used exclusively for receiving a call. According to an embodiment, the receiver may be formed integrally with or separately from the speaker. A sound output pattern of the sound output module **163** may be integrated with the display module **140**.

The camera module **171** may capture still images and moving images. According to an embodiment, the camera module **171** may include one or more lenses, an image sensor or an image signal processor. The camera module **171** may further include an infrared camera capable of determining a presence or an absence of a user, the user's location and the user's gaze.

The light module **172** may provide a light. The light module **172** may include a light emitting diode or a xenon lamp. The light module **172** may operate in conjunction with the camera module **171** or operate independently.

The communication module **173** may support establishment of a wired or wireless communication channel between the electronic apparatus **101** and the external electronic apparatus **102** and communication through the established communication channel. The communication module **173** may include one or both of a wireless communication module such as a cellular communication module, a short-distance wireless communication module, or a global navigation satellite system ("GNSS") communication module and a wired communication module such as a local area network ("LAN") communication module, or a power line communication module. The communication module **173** may communicate with the external electronic apparatus **102** through a short-range communication network such as Bluetooth, WiFi direct or infrared data association ("IrDA") or a long-distance communication network such as a cellular network, the Internet, or a computer network (e.g., LAN or WAN). The various types of communication modules **173** described above may be implemented as a single chip or may be implemented as separate chips.

The input module **130**, the sensor module **161** and the camera module **171** may be used to control the operation of the display module **140** in conjunction with the processor **110**.

The processor **110** outputs commands or data to the display module **140**, the sound output module **163**, the camera module **171** or the light module **172** based on the input data received from the input module **130**. In an embodiment, for example, the processor **110** may generate image data corresponding to input data applied through a mouse or an active pen, and output the generated image data to the display module **140** or the processor **110** may generate command data corresponding to the input data and output the generated command data to the camera module **171** or the light module **172**. When input data is not received from the input module **130** for a certain period of time, the processor **110** converts an operation mode of the electronic apparatus **101** into a low power mode or a sleep mode so that a power consumption of the electronic apparatus **101** may be reduced.

The processor **110** outputs commands or data to the display module **140**, the sound output module **163**, the camera module **171** or the light module **172** based on sensed data received from the sensor module **161**. In an embodiment, for example, the processor **110** may compare authentication data applied by the fingerprint sensor **161-1** with authentication data stored in the memory **120**, and then execute an application according to the comparison result. The processor **110** may execute commands or output corresponding image data to the display module **140** based on the sensed data sensed by the input sensor **161-2** or the digitizer **161-3**. When the sensor module **161** includes a temperature sensor, the processor **110** may receive temperature data for the temperature measured from the sensor module **161** and may further perform luminance correction on the image data based on the temperature data.

The processor **110** may receive determined data about the presence or the absence of the user, the user's location and the user's gaze from the camera module **171**. The processor **110** may further perform luminance correction on the image data based on the determined data. In an embodiment, for example, the processor **110**, which determines the presence or the absence of the user through an input from the camera module **171**, may display image data having the luminance corrected by the data converting circuit **112-2** or the gamma correction circuit **112-3** to the display module **140**.

Some of the above elements may be connected to each other through a communication method between peripheral devices such as a bus, a general purpose input/output ("GPIO"), a serial peripheral interface ("SPI"), a mobile industry processor interface ("MIPI"), or an ultra path interconnect ("UPI") link to exchange signals (e.g., commands or data) with each other. The processor **110** may communicate with the display module **140** through an agreed interface. In an embodiment, for example, the processor **110** may communicate with the display module **140** through any one of the above communication methods. The present invention may not be limited to the above communication methods.

The electronic apparatus **101** according to various embodiments disclosed in the disclosure may be various types of apparatuses. In an embodiment, for example, the electronic apparatus **101** may include at least one of a portable communication apparatus (e.g., a smart phone), a computer apparatus, a portable multimedia apparatus, a portable medical apparatus, a camera, a wearable device and a home appliance. The electronic apparatus **101** according to the embodiment of the disclosure may not be limited to the aforementioned apparatuses.

In an embodiment, for example, the display panel **100** of FIG. **1** may correspond to the display panel **141** of FIG. **20**. For example, the driving controller **200** of FIG. **1** may correspond to the controller of the auxiliary processor **112** of FIG. **20**. In an embodiment, for example, the gate driver **300** of FIG. **1** may correspond to the scan driver **142** of FIG. **20**. For example, the data driver **500** of FIG. **1** may correspond to the data driver **143** of FIG. **20**.

For example, the power voltage generator **600** of FIG. **1** may correspond to the power module **150** of FIG. **20**.

According to the embodiments of the display apparatus in the present invention, the constant current of the gamma reference voltage generator and the data driver may be reduced so that the power consumption of the display apparatus may be effectively reduced.

The foregoing is illustrative of the present invention and is not to be construed as limiting thereof. Although a few embodiments of the present invention have been described, those skilled in the art will readily appreciate that many modifications are possible in the embodiments without materially departing from the novel teachings and advantages of the present invention. Accordingly, all such modifications are intended to be included within the scope of the present invention as defined in the claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents but also equivalent structures. Therefore, it is to be understood that the foregoing is illustrative of the present invention and is not to be construed as limited to the specific embodiments disclosed, and that modifications to the disclosed embodiments, as well as other embodiments, are intended to be included within the scope of the appended claims. The present invention is defined by the following claims, with equivalents of the claims to be included therein.

What is claimed is:

1. A display apparatus comprising:

a gamma reference voltage generator configured to generate a gamma reference voltage;

a digital-to-analog converter configured to convert a data signal to a data voltage based on the gamma reference voltage;

an amplifier configured to receive the data voltage from the digital-to-analog converter and to output the data voltage to a data line; and

a bias voltage applier configured to output a first bias voltage to the gamma reference voltage generator and to output a second bias voltage to the amplifier,

wherein the bias voltage applier is turned off and does not supply the first bias voltage to the gamma reference voltage generator and the second bias voltage to the amplifier while input image data have a black image.

2. The display apparatus of claim **1**, wherein the gamma reference voltage generator and the amplifier are turned off while the input image data have the black image.

3. The display apparatus of claim **1**, further comprising:

a channel connected to the data line;

a first switch connected between an output terminal of the amplifier and the channel; and

a second switch connected between an input terminal of a hold voltage and the channel.

4. The display apparatus of claim **3**, wherein, in a normal mode in which the input image data include a grayscale value which is not the black image, the first switch is turned on and the second switch is turned off.

5. The display apparatus of claim **4**, wherein, in a black mode in which the input image data include only the black image, the first switch is turned off and the second switch is turned on.

6. The display apparatus of claim **3**, wherein the hold voltage is a high data power voltage generated by a power voltage generator.

7. A display apparatus comprising:

a display panel including a data line extending in a second direction;

a channel connected to the data line;

an amplifier configured to output a data voltage;

a first switch connected between an output terminal of the amplifier and the channel; and

a second switch connected between an input terminal of a hold voltage and the channel,

wherein when input image data have a first black pattern extending in a first direction perpendicular to the second direction, the first switch is turned off and the second switch is turned on such that the input terminal of the hold voltage supplies the hold voltage, which is a predetermined direct current (DC) voltage, to the date line through the channel during a black period corresponding to the first black pattern, and the first switch is turned on and the second switch is turned off during a normal period not corresponding to the first black pattern.

8. The display apparatus of claim **7**, wherein a data signal includes a line configuration and line data, and

wherein the line configuration includes a setting signal representing the black period and the normal period.

9. The display apparatus of claim **8**, wherein, when proceeding from the black period to the normal period, the setting signal is determined based on a wakeup time.

10. The display apparatus of claim **7**, further comprising:

a gamma reference voltage generator configured to generate a gamma reference voltage;

a digital-to-analog converter configured to convert a data signal to the data voltage based on the gamma reference voltage; and

a bias voltage applier configured to output a first bias voltage to the gamma reference voltage generator and to output a second bias voltage to the amplifier.

11. The display apparatus of claim **10**, wherein the bias voltage applier is turned off during the black period.

12. The display apparatus of claim **11**, wherein the gamma reference voltage generator and the amplifier are turned off during the black period.

13. A display apparatus comprising:

a display panel including a first display block and a second display block, which extend in an extension direction of a data line, wherein the data line includes a first data line and a second data line;

a first channel connected to the first data line of the first display block;

a first amplifier configured to output a first data voltage to the first channel;

A-1 switch connected between an output terminal of the first amplifier and the first channel;

B-1 switch connected between an input terminal of a hold voltage and the first channel;

a second channel connected to the second data line of the second display block;

a second amplifier configured to output a second data voltage to the second channel;

A-2 switch connected between an output terminal of the second amplifier and the second channel; and

B-2 switch connected between an input terminal of the hold voltage and the second channel,

wherein the A-1 switch and the B-1 switch are controlled independently from the A-2 switch and the B-2 switch,

wherein the input terminal of the hold voltage supplies the hold voltage, which is a predetermined direct current (DC) voltage, to the first data line through the first channel when the B-1 switch is turned on and to the second data line through the second channel when the B-2 switch is turned on.

14. The display apparatus of claim **13**, wherein, when the data line extends in a second direction, input image data have a second black pattern extending in the second direction and the second black pattern overlaps an entirety of the first display block, the A-1 switch is turned off and the B-1 switch is turned on.

15. The display apparatus of claim **13**, wherein, when the data line extends in a second direction, input image data have a second black pattern extending in the second direction and the second black pattern partially overlaps the first display block or does not overlap the first display block, the A-1 switch is turned on and the B-1 switch is turned off.

16. The display apparatus of claim **13**, wherein, when the data line extends in a second direction, input image data have a second black pattern extending in the second direction, the second black pattern overlaps an entirety of the first display block and the second black pattern partially overlaps the second display block or does not overlap the second display block, the A-1 switch is turned off, the B-1 switch is turned on, the A-2 switch is turned on and the B-2 switch is turned off.

17. The display apparatus of claim **13**, wherein, when the data line extends in a second direction, input image data have a third black pattern extending in a first direction perpendicular to the second direction in the first display block, the A-1 switch is turned off and the B-1 switch is turned on during a black period corresponding to the third black pattern and the A-1 switch is turned on and the B-1 switch is turned off during a normal period not corresponding to the third black pattern.

18. The display apparatus of claim **17**, wherein a data signal includes a line configuration and line data,

wherein the line configuration includes a first display block setting signal representing the black period of the first display block and the normal period of the first display block and a second display block setting signal representing the black period of the second display block and the normal period of the second display block.

19. The display apparatus of claim **13**, further comprising:

a gamma reference voltage generator configured to generate a gamma reference voltage;

a digital-to-analog converter configured to convert a data signal to a data voltage based on the gamma reference voltage; and

a bias voltage applier configured to output a first bias voltage to the gamma reference voltage generator and to output a second bias voltage to the first amplifier and the second amplifier,

wherein the data voltage includes the first data voltage and the second data voltage.

20. The display apparatus of claim **19**, wherein while input image data have a black image for all display blocks in the display panel, the bias voltage applier is turned off.

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